

ASSESSMENT OF MEDITERRANEAN SWORDFISH (*Xiphias gladius*) STATUS USING BAYESIAN STATE-SPACE SURPLUS PRODUCTION MODEL JABBA

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SUMMARY

Bayesian State-Space Surplus Production Models, implemented via the 'JABBA' R package, were applied to catch and CPUE data for Mediterranean swordfish (Xiphias gladius). Three different configurations were implemented: a Schaefer formulation with a pre-existing prior distribution for r , a Pella-Tomlinson formulation informed by priors derived from an Age-Structured Equilibrium Model and a modification of the latter that estimates the shape parameter. Continuity runs were evaluated in order to inspect the effect of the data update with respect to 2020 Stock Assessment. Several sensitivity runs were evaluated to all of the 'main' models in order to test the robustness of model results to alternative configurations and data assumptions.

RESUMEN

RÉSUMÉ

KEYWORDS

Xiphias gladius; Mediterranean; JABBA; stock status; production models

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Introduction

Mediterranean swordfish fishery is managed and assessed by the International Commission for the Conservation of Atlantic Tunas (ICCAT). Assessments of the stock status of the species are conducted periodically.

In 2017 a fifteen year multi-annual Recovery plan was adopted for Mediterranean swordfish (ICCAT Recommendation 16-05), following the outcome from the stock assessment conducted by SCRS in 2016, which showed an overfished status as well as overfishing. Several conservation and control measures were recommended in the Recovery Plan, with the goal of achieving B_{MSY} with at least 60% probability. The conservation measures included total allowable catch (TAC), capacity limitations, a closed fishing season, change in the minimum landing size (MLS), limits on the technical specifications of the fishing gear and, finally information on recreational fisheries.

The most recent stock assessment was conducted in 2020 (Anon., 2020), when several models were attempted, including Extended Survivors Analysis (XSA; Kell et al., 2007), Assessment for All (a4a; Jardim et al., 2015) and Bayesian state-space production model (JABBA; Winker et al., 2018). Scientific advice was based on the combination of two JABBA models with different production functions, Schaefer and Pella-Tomlinson model types. For the former the r prior distribution was derived from a Leslie matrix model approach using Monte-Carlo simulations (Anon., 2015), while for the later an Age-Structured Equilibrium Model (ASEM) approach with Monte-Carlo simulations (Winker et al., 2020a) was used to develop an r prior distribution with an associated shape parameter m of a Pella-Tomlinson production function. The Mediterranean swordfish biomass (B_{2018}) was below B_{MSY} , while fishing mortality rate was close to the sustainable level (F_{MSY}) resulting in a stock status being overfished but overfishing was not occurring.

This document's goal is to show the results for candidate 2026 stock assessment models for the Mediterranean swordfish stock using the Bayesian state-space production model platform JABBA, using updated series of catch and CPUE up to 2024.

1. Data and methods

1.1 Fishery data

ICCAT Secretariat provided the available catch data from year 1950 to 2024 (**Figure 1**). Catches were split in landings (1950-2024) and dead discards (2004-2024). Estimation of discards from the main longline fleets in the Mediterranean was provided by ICCAT Secretariat following Ortiz and Kimoto, 2026. In addition, EU-Spain provided an estimation of discards from the longline fleet operating in the western Mediterranean (Rueda et al., 2026).

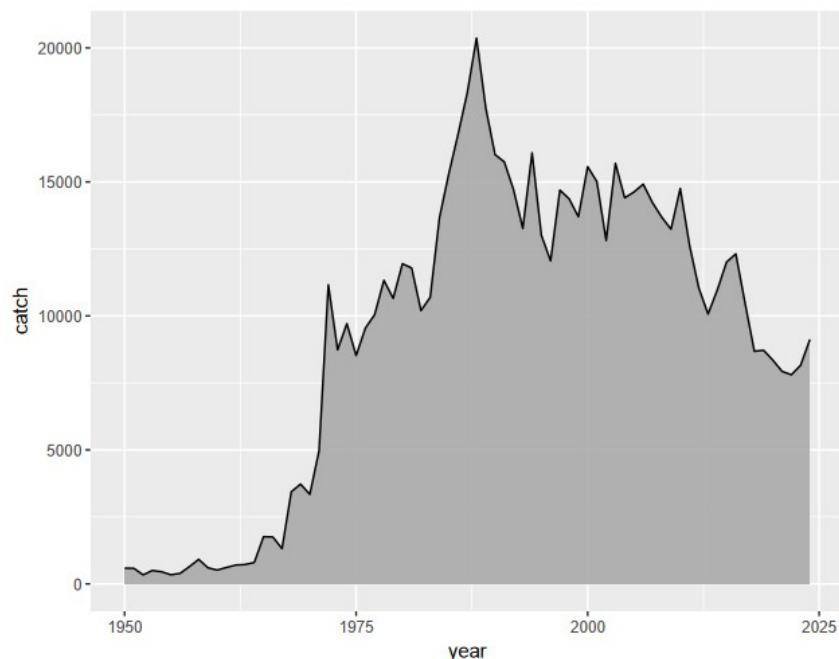


Figure 1. Total catches (tons) of swordfish in the Mediterranean from 1950 to 2024.

Standardized catch (in weight) per unit of effort (CPUE) series were used as relative indices of abundance. Four longline standardized CPUE abundance indices were provided (**Figure 2**): Greece (from 1987 to 2024; Tserpes and Peristeraki, 2026), Spain (from 1988 to 2024; Saber et al. 2026), Liguria-Italy (from 2010 to 2023; Garibaldi et al. 2026) and Morocco (from 2012 to 2024; Layachi et al. 2026).

In addition, other abundance indices were provided for inclusion in sensitivity analyses: a Sicilian gillnet index (from 1990 to 2009; Tserpes et al. 2010) and a historical Ligurian longline index (from 1991 to 2009; Garibaldi and Tserpes, 2014).

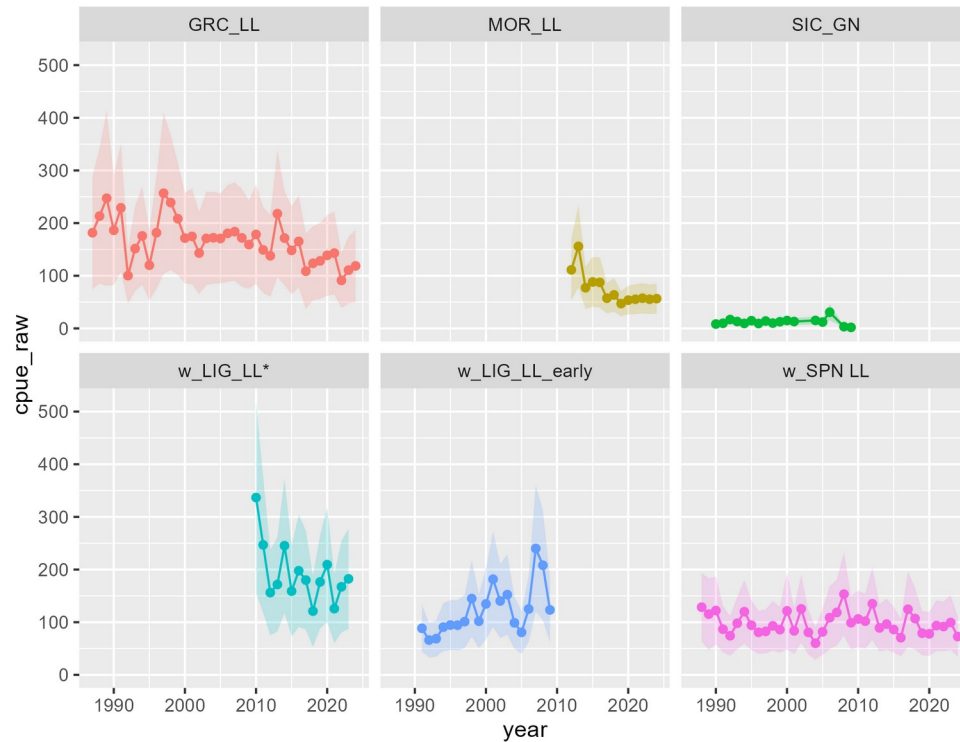


Figure 2. Time series of relative indices of abundance. GRC_LL: Greece longline; MOR_LL: Morocco longline; SIC_GN: Sicilian gillnet; w_LIG_LL: Ligurian longline; w_LIG_LL_early: historical Liguria longline; w_SPN_LL: Spain longline.

Coefficients of variation (CVs) for each year from the standardized CPUEs were adjusted for each index to have a minimum value of 0.25 while keeping the inter-annual variability from the original CVs reported by the index providers. Specifically, for each CPUE series, a non-negative constant value was added to the whole series so that its minimum value was 0.25. This modification was agreed by the Swordfish species group during the Swordfish Data Preparatory Meeting of March 2026 (Anon., 2026).

1.2 Biological data

In addition to fisheries data, life history parameters (**Table 1**) on Mediterranean swordfish were used to estimate the r prior distribution for one of the three main candidate models presented in this document (see 1.3 Model specifications). The same parameters were used in the 2020 assessment with the exception of length at first maturity (L_{50}) value, which was updated in Borrego et al., 2026.

Table 1. Adopted parameters to be used in the assessment

Parameter	Mean	CV	Distribution	Description	Source
M	0.206	0.25	lognormal	Natural mortality (1/year)	McAllister (2014)
L_{inf}	238.58	0.1	lognormal	Von Bertalanffy asymptotic length	Mean: ICCAT Manual CV: Working Group
K	0.185	0.1	normal	Von Bertalanffy growth parameter	Mean: ICCAT Manual CV: Working Group
t_0	-1.404	0.2	normal	Von Bertalanffy age at	Mean: ICCAT Manual

				zero length	CV: Working Group
a	9.61E-06	-	exponential	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
b	3.059	-	non lognormal	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
L ₅₀	136.5	0.2	lognormal	Length at 50% maturity	Mean: SCRS/2026/055 Manual CV: McAllister (2014)
d	0.2	0.2	lognormal	Parameter of the logistic maturity ogive	Working Group
h	0.83	0.14	beta	Steepness $h=0.2 + 0.8$ Beta (5.86. 1.59)	McAllister (2014)

1.3 Model specifications

The R (R Core Team 2026) package 'JABBA' v.2.3.5 (Winker et al. 2026) was used to configure and fit the models. Following the default configuration for JABBA models, the unfished biomass (K) was assigned a vaguely informative lognormal prior: centered at eight times the maximum historical catch, with a coefficient of variation (CV) of 1 (100%). For the initial depletion (B_{1950}/K), a lognormal prior with mean 0.95 and CV equal to 0.1 (10%) was used. The random catch variability on log scale was set to a value of 0.2. Catchability parameters associated to each one to the CPUE indices were formulated as uniform priors and observation variances were estimated by index assuming inverse-gamma priors. A minimum fixed observation error equal to 0.001 was used. The state-space formulation allows annual process error deviations around the deterministic surplus production. The process error variance was estimated using an non-informative inverse-gamma prior with both scaling parameters set to 0.001.

The analyses included three different categories of model runs:

- I. *Continuity runs*: two Continuity runs were conducted using updated series (up to 2024) on catch and relative indices of abundance (see section 1.1 Fishery data), one for the 2020 Schaefer Reference run and another for the 2020 ASEM run. The values of the parameters used as well as model specifications can be found in Winker et al., 2020b and Anom., 2020.
- II. *Main runs*: three Main runs were conducted using both Schaefer and Pella-Tollimson production functions. In all cases the above-mentioned model specifications were used.
 - a. Schaefer: For the Schaefer model, the r prior distribution from the 2020 assessment was used (Winker et al. 2020b, Anom, 2020).
 - b. Pella: For the Pella-Tollimson Main run new r prior distribution and the associated parameter (m) were calculated using an Age-Structured Equilibrium Model (ASEM) approach (Winker et al., 2020a). The values of the parameters used to run ASEM are shown in **Table 1**. In addition to the updated L₅₀ value with respect to the 2020 ASEM run, the following ASEM parameters were updated, based on inter-sessional discussions: (i) slope parameter of L₅₀ (dL_m) equal to 1/0.2; (ii) maximum age equal to 15 years; (iii) sequence of range of F values for iterative MSY estimation from 0.01 to 0.9 each 0.002; (iv) length at 50% retention (SL_{50}) equal to 100 cm based on the analysis from the cumulative frequency distribution from the Spanish measured lengths. The r prior specification and the B_{MSY}/K value that were obtained from the ASEM simulations is shown in **Table 2**. However, following the same criterium used in the 2020 stock assessment, the final value for the CV of the prior of r was set to 0.2.
 - c. Pella-m: The third Main model is an adaptation of the Pella ASEM model where the shape parameter m estimated within the model. The prior specification for r was the same median as the Pella model and the CV on the shape parameter was set to 0.3.

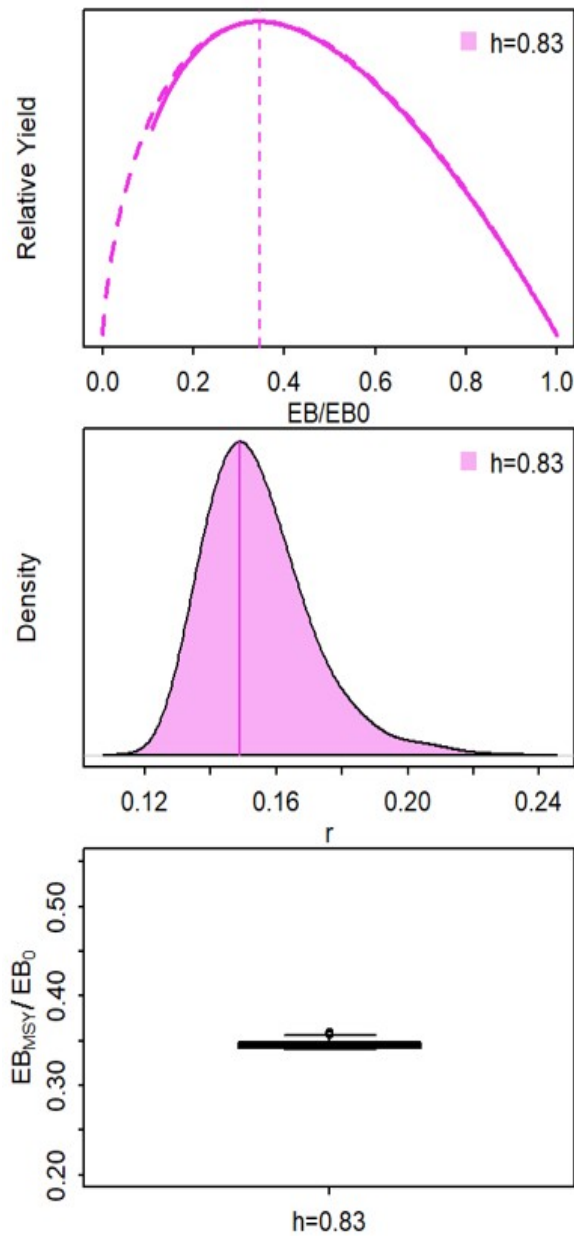


Figure 3. ASEM simulation results: Relative production curve (top panel), marginal r distribution (middle panel) and boxplot of B_{MSY}/K (bottom panel)

Table 2. Results of the ASEM model parameter values

μr	0.154
$CV r$	0.096
B_{MSY} / K	0.35

III. *Sensitivity runs:* Five sensitivity scenarios were considered for each of the Main runs.

- (i) Jackknife index analysis consisting on re-fitting the Main run excluding one of the four abundance indices used in the Main run at a time.
- (ii) under-reporting catches using alternative catches scenarios by scaling the catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3 and re-fitting the Main model with the inflated historical catches.
- (iii) set the initial biomass depletion (B_{1950}/K) prior mean to 0.85.
- (iv) a run with no random variation on catches.
- (v) a run using the historical abundance indices 'w_LIG_LL_early' and 'SIC_GN' (**Figure 2**).

1.4 Diagnostics

To evaluate CPUE fits JABBA residual plots were used to compare predicted versus observed values. JABBA provides (i) colour-coded lognormal CPUE residuals by index and year, (ii) boxplots of the median and quantiles of the residuals by year and index and (iii) loess-smoother through each index residuals by year. It also provides a root mean squared error (RMSE) statistic.

In addition, a retrospective analysis was used to check for bias in stock status estimates by removing one year of data at a time sequentially up to five years and re-fitting the Main model runs to the data. Outputs for the quantities of interest were compared between each model run removing data and the Main model run fitted to the whole time series. JABBA provides the Mohn's rho (ρ) statistic to compare the bias between the Main model with all data and the re-fitted models with sequentially removed data.

2. Results

2.1 Continuity runs

The Continuity runs showed substantial differences in the trajectories of the B/B_{MSY} and F/F_{MSY} (**Figure 4**) with respect to those in the Reference and ASEM runs in 2020 (Anom, 2020). The probability of being in the green quadrant (B) of the Kobe plot changed from 10.4% in 2020 to 47.2% in 2026 (Schaefer), and from 21.1% in 2020 to 50.7% in 2026 (Pella-Tollimson).

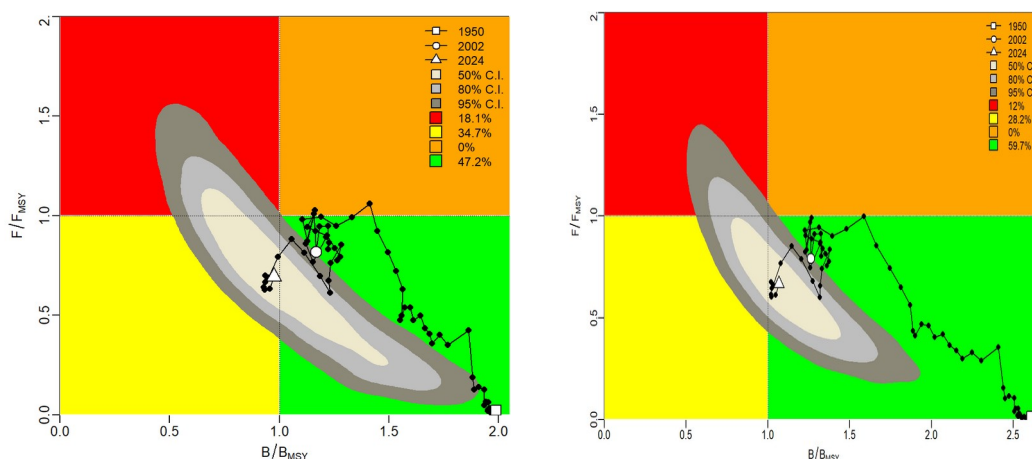


Figure 4. Kobe phase plots for the 2026 Continuity runs for Schaefer (left) and Pella-Tollimson (right) models.

To further explore these differences several sensitivity runs were conducted on both continuity runs (Schaefer and Pella), these being: (i) Jackknife index analysis, (ii) Modifying input data, setting a fixed CV equal to 0.2 for all years and all indices (similar to the 2020 assessment), (iii) fitting models up to 2018 (last year of data of the 2020 assessment) but using the relative abundance indices provided for the present assessment and, finally (iv) re-fitting the continuity runs but using the old relative abundance indices, one at a time.

These results (not all shown here) indicated that the change in estimated trajectories of B/B_{MSY} and F/F_{MSY} in the 2026 Continuity runs is potentially driven by the updated Greek longline relative abundance index (**Figure 4**). When considering all the sensitivity runs explored for the Continuity models, only the one using the 'old' Greek index (the index used in the 2020 assessment) — applied to both the Schaefer and Pella-Tomlinson formulations from 2020 (**Figures 5, 6**) — reproduced dynamics similar to those observed in the 2020 assessment over the common time period (1950 – 2018).

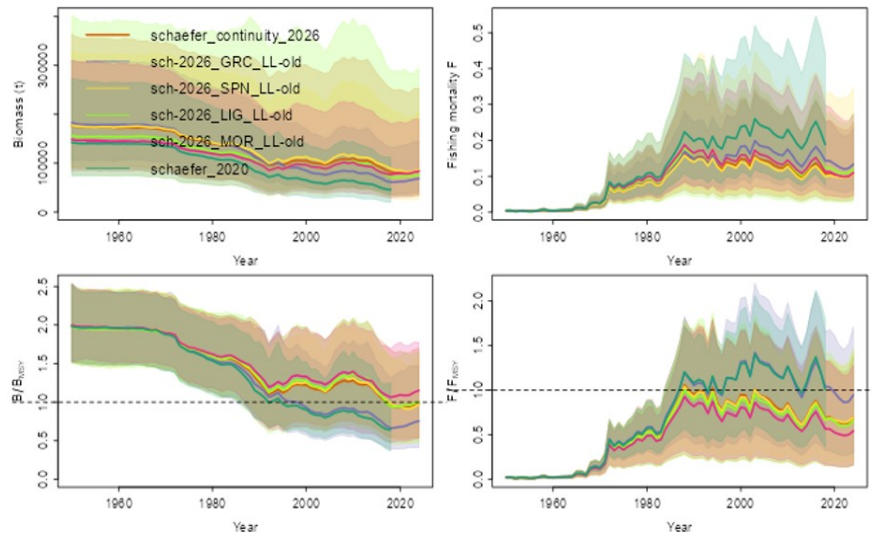


Figure 5. Output of the sensitivity run using the 2020 assessment indices of abundance ('old') on the Schaefer Continuity run. Schaefer 2020: results from the 2020 assessment. Schaefer_continuity_2026: 2026 continuity run; sch-2026_XX_LL-old: 2026 continuity runs using three updated indices of abundance (present assessment) and an old index (2020 assessment) at a time.

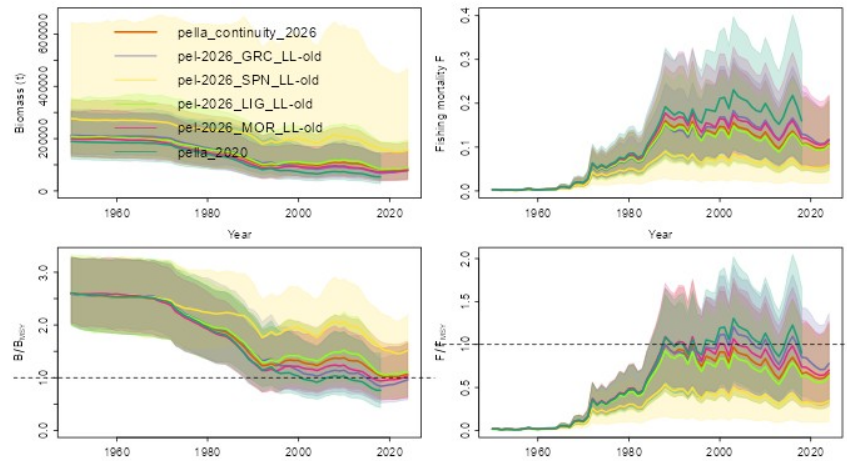


Figure 6. Output of the sensitivity run using the 2020 assessment indices of abundance ('old') on the Pella-Tollimson Continuity run. Pella 2020: results from the 2020 assessment. Pella_continuity_2026: 2026 continuity run; pel-2026_XX_LL-old: 2026 continuity runs using three updated indices of abundance (present assessment) and an old index (2020 assessment) at a time.

Visual inspection of the comparison between input abundance indices (**Figure 7**) shows a contrasting phase in the early years of the Greek index of abundance, the 2020 index shows higher values in the earlier period (1987-1999) and lower values from 2020 to 2018 after a steep decline in 1998-2000. In contrast, the updated index is more stable and does not show such differences between the earlier period and the most recent years. Some discrepancies are also observed in the updated indices of LIG_LL and MOR_LL, whereas the Spanish updated index is the most similar to the previous one.

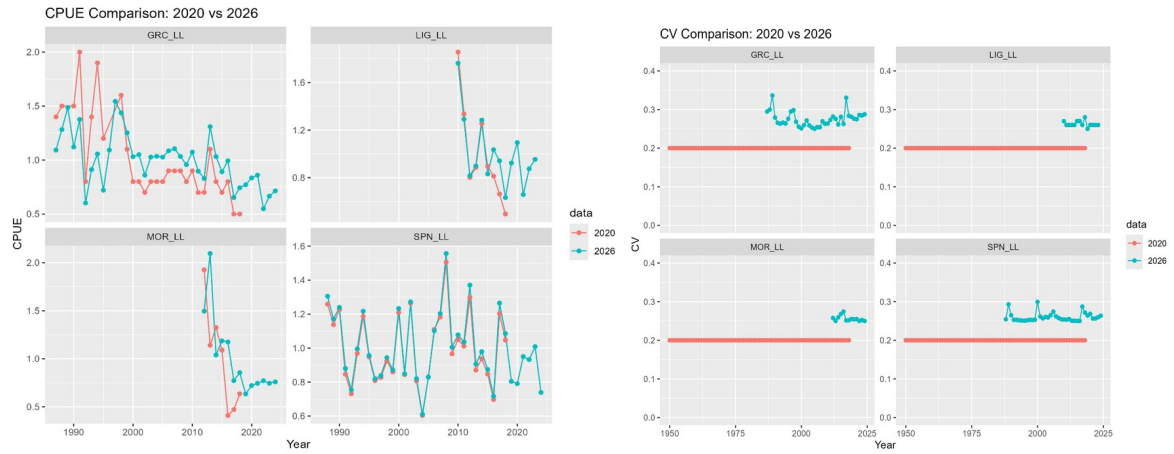


Figure 7. Comparison of indices of abundance timeseries. Left panel: CPUE timeseries . Right panel: CV timeseries.

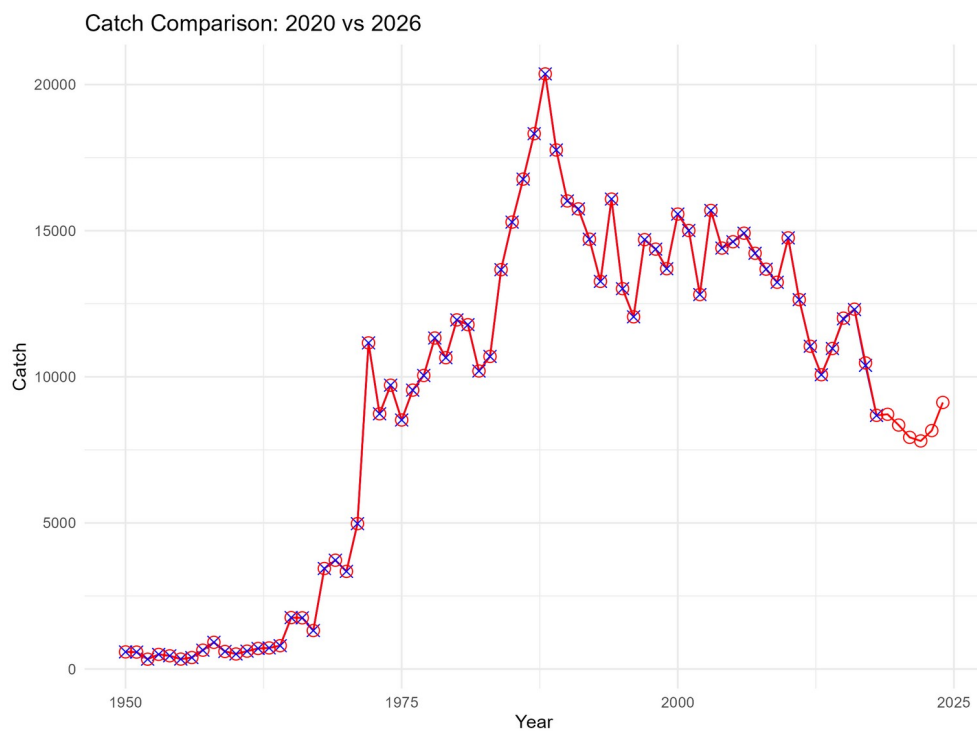


Figure 8. Comparison of input catch timeseries used in the 2020 assessment (blue crosses) and the one from 2026 (red circles).

2.2 Main runs

For each of the Main model runs, posterior distributions for all estimated quantities were obtained via Markov Chain Monte Carlo (MCMC) simulation using JAGS, called from R through the 'r2jags' interface. 2 MCMC chains were run, each for 100,000 iterations, discarding the first 12,000 iterations as burn-in and thinning the remaining chain by a factor of 5, yielding 35,200 posterior draws used for inference.

The model fits to the standardized CPUE series are presented in **Figure 9** . Overall, the three main models provided a satisfactory fit to the CPUE data.

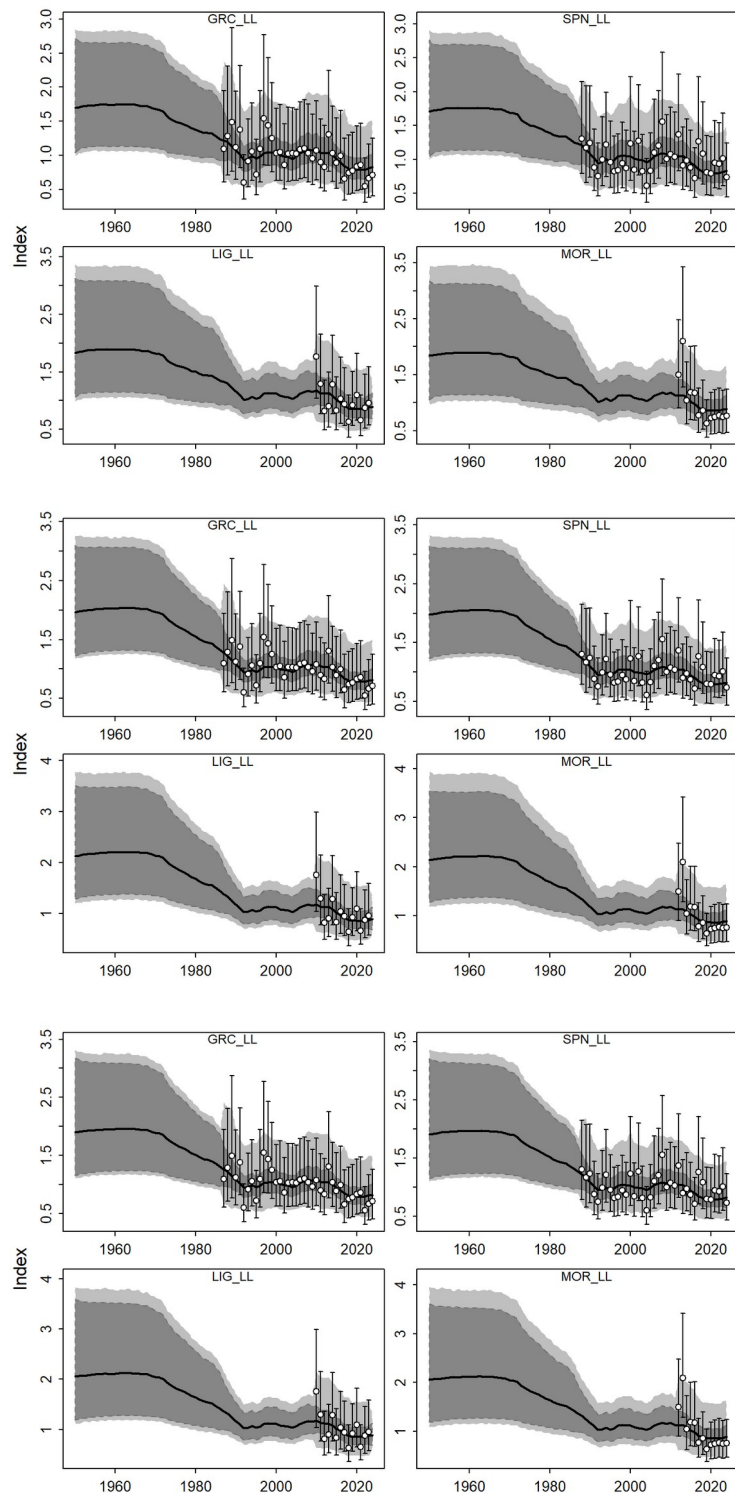


Figure 9. Observed CPUEs (circles) with 95% confidence interval (CI) error bars and predicted CPUE (solid line) from the JABBA model. Top panel: Schaefer Main run; Middle panel: Pella-Tollimson ASEM Main run; Bottom panel: Pella-est_m Main run. Dark grey shaded areas show the 95% credibility intervals from the expected CPUE and light grey shaded area shows the 95% posterior predictive distribution intervals.

Marginal posterior distributions together with their corresponding priors are displayed in **Figure 10**. Posterior median values and 95% confidence intervals can be found in **Table 2**.

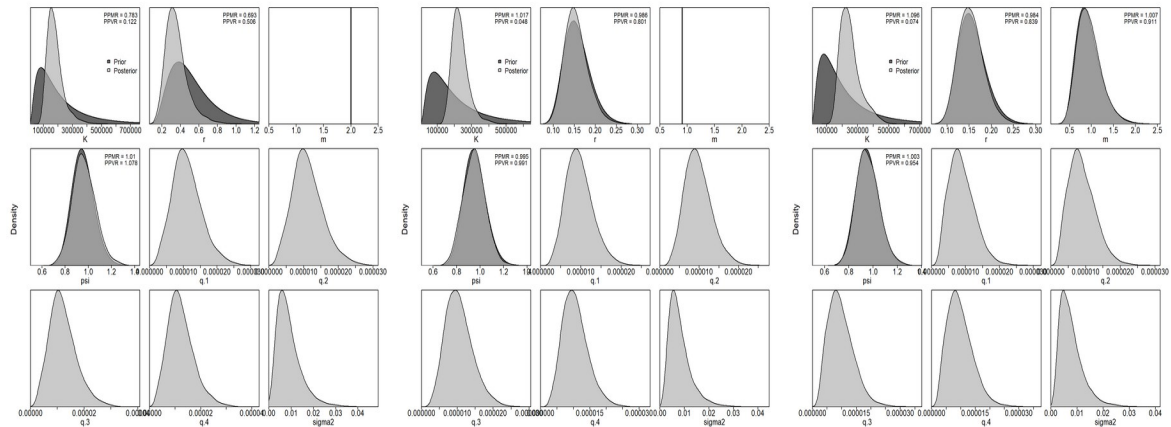


Figure 10. Prior and posterior distributions for the parameters. PPRM: Posterior to Prior Ratio of Means; PPRV: Posterior to Prior Ratio of Variances. Left panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Right panel: Pella-est_m Main run.

Table 3. Posterior median values and 95% confidence intervals for the three Main model runs.

Schaefer					
	Median	LCI	UCI	Geweke.p	Heidel.p
K	1.685092e+05	9.963645e+04	3.380386e+05	0.637	0.722
r	3.390181e-01	1.792374e-01	6.794201e-01	0.142	0.328
q.1	1.060000e-05	3.700000e-06	2.120000e-05	0.931	0.917
q.2	1.070000e-05	3.700000e-06	2.150000e-05	0.895	0.914
q.3	1.150000e-05	3.800000e-06	2.410000e-05	0.985	0.931
q.4	1.150000e-05	3.900000e-06	2.430000e-05	0.904	0.886
φ	9.578258e-01	7.806876e-01	1.175400e+00	0.834	0.101
sigma2	7.786900e-03	2.059500e-03	2.345230e-02	0.823	0.303
tau2.1	2.917600e-03	4.221000e-04	2.000470e-02	0.979	0.198
tau2.2	3.060400e-03	4.659000e-04	2.228390e-02	0.641	0.336
tau2.3	6.551500e-03	7.049000e-04	6.632640e-02	0.251	0.306
tau2.4	8.858500e-03	5.540000e-04	1.013433e-01	0.738	0.711
m	2.000000e+00	2.000000e+00	2.000000e+00	NaN	NA
Pella ASEM					
	Median	LCI	UCI	Geweke.p	Heidel.p
K	2.263195e+05	1.589714e+05	3.621131e+05	0.378	0.392
r	1.516330e-01	1.069767e-01	2.177541e-01	0.764	0.359
q.1	9.300000e-06	4.000000e-06	1.710000e-05	0.087	0.114
q.2	9.400000e-06	4.000000e-06	1.740000e-05	0.085	0.144
q.3	1.010000e-05	4.200000e-06	1.930000e-05	0.048	0.091
q.4	1.010000e-05	4.300000e-06	1.940000e-05	0.171	0.224
φ	9.484221e-01	7.749153e-01	1.140076e+00	0.563	0.118
sigma2	7.063600e-03	2.350400e-03	2.026990e-02	0.125	0.119

Pella ASEM					
tau2.1	3.121100e-03	4.749000e-04	2.008580e-02	0.907	0.666
tau2.2	3.179100e-03	4.915000e-04	2.222130e-02	0.676	0.800
tau2.3	6.194900e-03	5.820000e-04	6.731020e-02	0.409	0.406
tau2.4	8.766400e-03	5.612000e-04	1.005923e-01	0.513	0.283
m	9.070000e-01	9.070000e-01	9.070000e-01	NaN	NA

Pella-Est_m					
	Median	LCI	UCI	Geweke.p	Heidel.p
K	2.395624e+05	1.586851e+05	4.187666e+05	0.827	0.938
r	1.517561e-01	1.054065e-01	2.178550e-01	0.997	0.790
q.1	8.400000e-06	3.200000e-06	1.760000e-05	0.986	0.896
q.2	8.500000e-06	3.200000e-06	1.780000e-05	0.853	0.833
q.3	9.100000e-06	3.300000e-06	1.980000e-05	0.739	0.831
q.4	9.100000e-06	3.300000e-06	1.990000e-05	0.966	0.837
ϕ	9.508470e-01	7.900412e-01	1.158195e+00	0.950	0.529
sigma2	6.774900e-03	2.175600e-03	1.962540e-02	0.207	0.449
tau2.1	2.936400e-03	4.385000e-04	2.026450e-02	0.290	0.866
tau2.2	3.211100e-03	4.486000e-04	2.223880e-02	0.714	0.514
tau2.3	5.867300e-03	6.463000e-04	6.399070e-02	0.000	0.749
tau2.4	9.217700e-03	6.326000e-04	1.034056e-01	0.339	0.155
m	9.172791e-01	5.190161e-01	1.603323e+00	0.682	0.791

The posterior median of the intrinsic population growth rate (r) was estimated at 0.339 for the Schaefer model, at 0.151 for the Pella-ASEM model, and at 0.151 for the Pella-m model (**Table 3**). The relatively small prior-to-posterior median ratio (PPMR) indicate agreement between priors and posteriors, with substantial overlap between their distributions.

The posterior median estimate of carrying capacity (K) was 168,1509 metric tons for the Schaefer model, 226,319 for the Pella-ASEM, and 239,562 metric tons for the Pella-est model. Posterior distributions for K were noticeably narrower than their corresponding priors, and the resulting low PPVR values indicate that the input data were highly informative about this parameter in both models (**Figure 10**). In contrast, the posterior distributions for the initial depletion parameter (ϕ) were very similar to their priors, with both PPMR and PPVR remaining close to 1, suggesting that estimates of initial depletion were driven largely by the prior assumptions rather than by the data.

Posterior summaries for model parameters and key management quantities are provided in **Table 4**. Estimated MSY values were similar across models, with posterior medians of 13,723 metric tons for the Schaefer model, 13,081 metric tons for the Pella-ASEM model and 13,414 for the Pella-est_m model. The corresponding posterior median estimates of B_{MSY} were 84,255 metric tons for the Schaefer model, 79,229 for the Pella-ASEM and, finally 84,495 metric tons for the Pella-est-m (**Table 4**). Estimates of F_{MSY} were also comparable, with values around 0.16 for the three models.

Table 4. Posterior median values for key model parameters and reference quantities.

Model	quant	K	B_{MSY}	F_{MSY}	MSY
Schaefer	median	1.685092e+05	8.425462e+04	0.1695091	1.372280e+04
Pella ASEM	median	2.263195e+05	79229.408255	0.1671809	1.308110e+04
Pella-Est_m	median	2.395624e+05	8.449485e+04	0.1654880	1.341402e+04

All of the main models showed similar temporal trajectories for the median estimates of B , F , B/B_{MSY} , F/F_{MSY} , B/B_0 (**Figure 11, Tables 5, 6 and 7**), but there were differences in the scale among models, specially between the Schaefer Main run with respect to the Pella-ASEM and Pella-est_m runs. In all cases the biomass trajectory (B/B_{MSY}) exhibited an overall decline up to the 90s, followed by a more stable trend with another decline in the decade of the 2010s. Biomass was estimated to have fallen below the MSY reference point only under the Schaefer model during in the mid-2010s, whereas according to both Pella models this threshold was never crossed.

The trajectory of F/F_{MSY} increased gradually from the late 60s until the late 80s, following a moderate decline overall with noticeable fluctuations during the period 1990 – 2024 (**Figure 11**). F was estimated to reach values higher than F_{MSY} only in the Schaefer model.

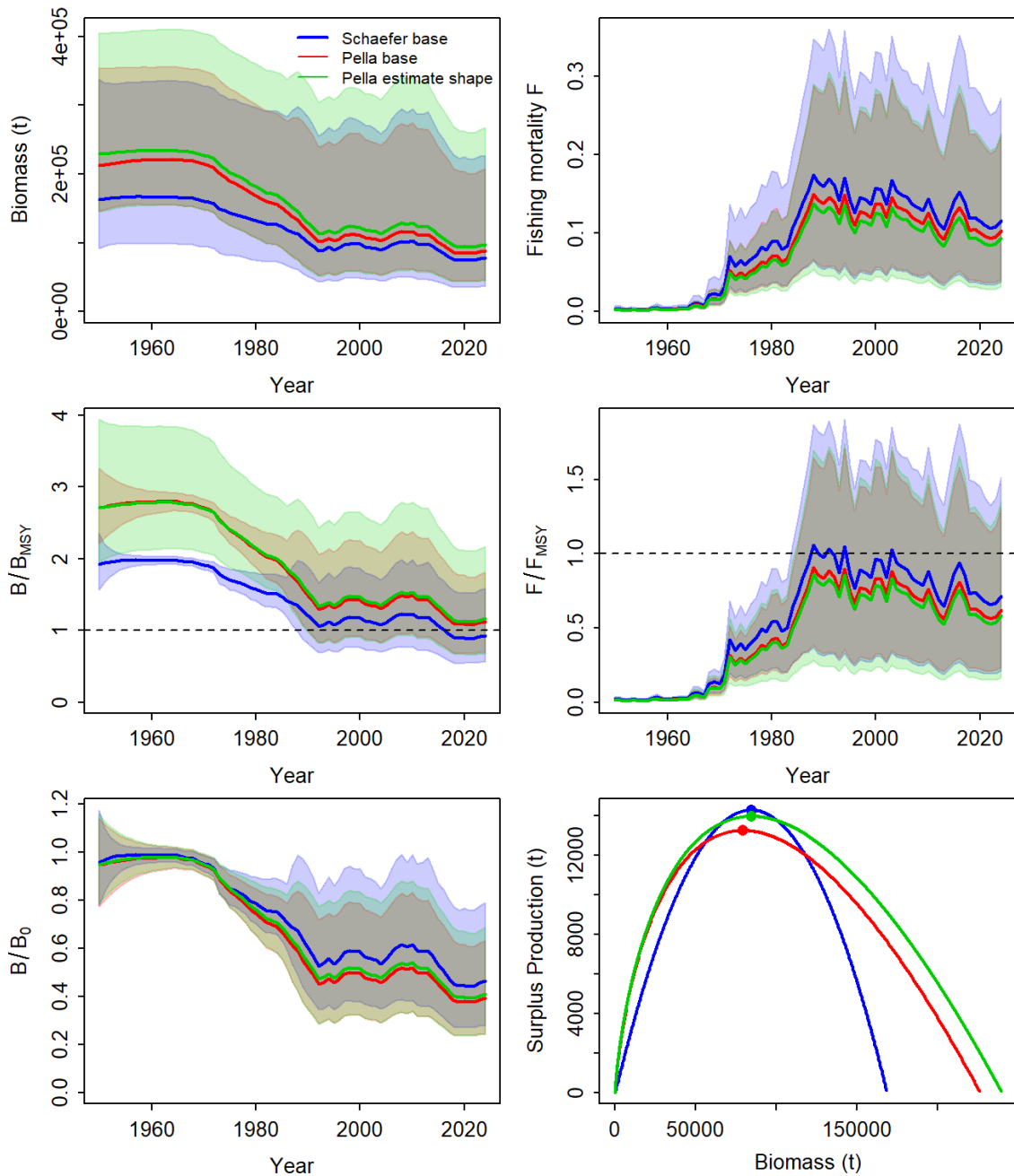


Figure 11. Trends in B and F (upper panels), B/B_{MSY} and F/F_{MSY} (middle panels), B/B_0 (left bottom panel) and surplus production curve (right bottom panel) for the three Main runs.

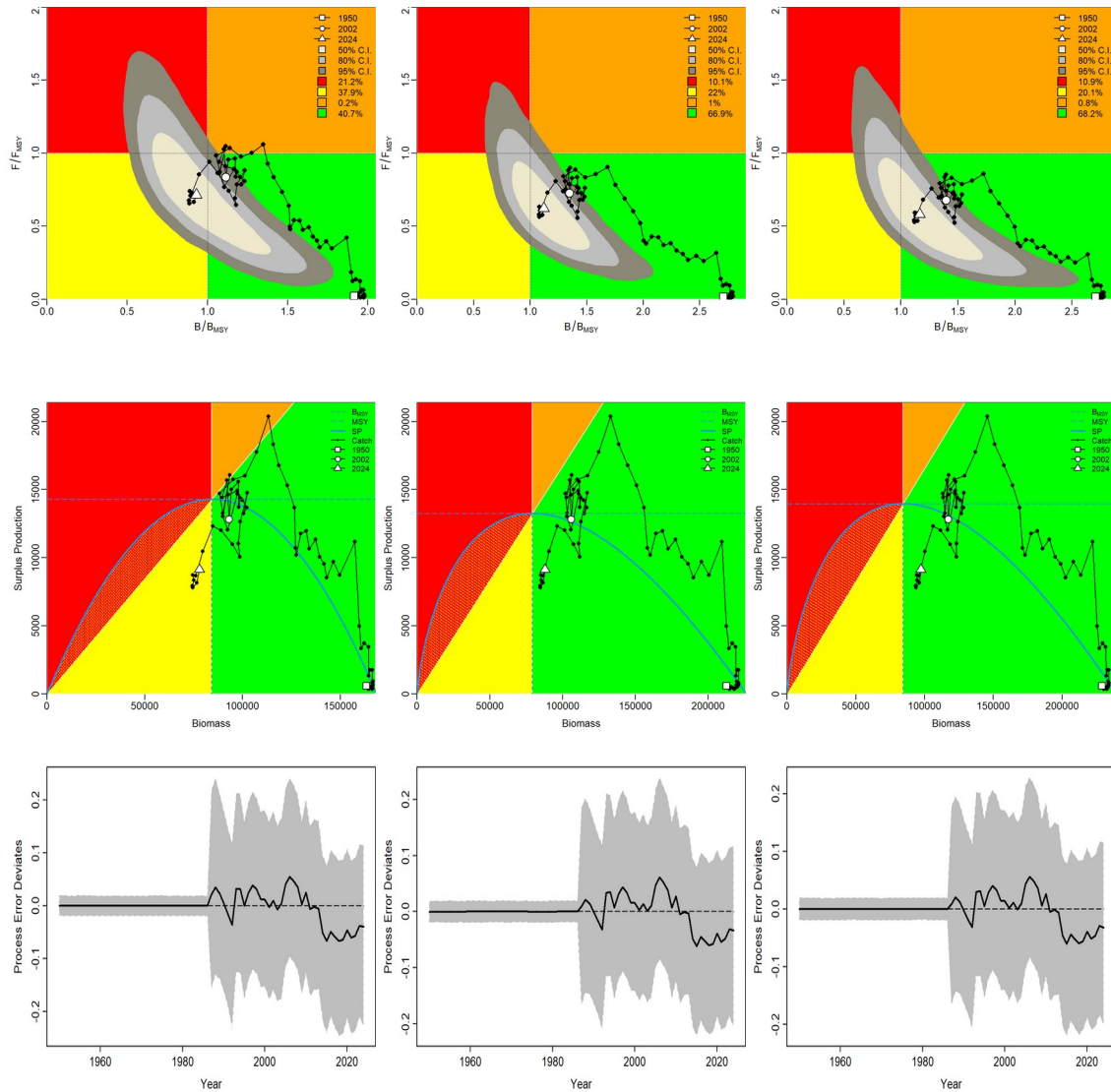


Figure 12. Top panels: Kobe phase plot for the three main model runs (left: Schaefer; Center: Pella-ASEM; right: Pella-m). Middle panels: JABBA surplus production phase plot for the three main model runs (left: Schaefer; Center: Pella-ASEM; right: Pella-m). Lower panel: process error deviations for the three main model runs (left: Schaefer; Center: Pella-ASEM; right: Pella-m).

Table 5. Estimates of biomass (B), fishing mortality (F), biomass relative to B_{MSY} , and fishing mortality relative to F_{MSY} between 1950 and 2024 in the form of joint MCMC posteriors of JABBA model runs for the Schaefer Main run.

Year	B	F	BB_{MSY}	FF_{MSY}	BB_0	procB	SPT
1950	163312.8114	0.0036	1.9161	0.0214	0.9580	0.0001	0.0000
1951	163611.9694	0.0035	1.9399	0.0212	0.9699	0.0001	2211.3332
1952	164459.6884	0.0020	1.9554	0.0122	0.9777	0.0001	1627.7263
1953	165287.6163	0.0030	1.9673	0.0181	0.9837	0.0000	1269.2209
1954	165870.3766	0.0027	1.9714	0.0162	0.9857	0.0000	968.1291
1955	166335.7286	0.0020	1.9749	0.0122	0.9875	0.0001	857.7335
1956	166563.6129	0.0023	1.9785	0.0140	0.9893	0.0000	745.5909
1957	166840.0389	0.0038	1.9801	0.0230	0.9900	0.0000	591.3086
1958	166803.2366	0.0055	1.9784	0.0326	0.9892	-0.0001	551.0176
1959	166409.7671	0.0036	1.9740	0.0214	0.9870	0.0001	610.7882
1960	166467.3256	0.0031	1.9757	0.0184	0.9878	0.0000	728.2552
1961	166611.5179	0.0037	1.9774	0.0219	0.9887	-0.0001	662.3253
1962	166526.5404	0.0042	1.9771	0.0251	0.9885	-0.0002	594.2995

1963	166512.4009	0.0043	1.9759	0.0258	0.9880	0.0000	622.1586
1964	166390.0951	0.0048	1.9754	0.0285	0.9877	0.0001	669.3990
1965	166307.5787	0.0106	1.9739	0.0629	0.9869	0.0000	679.7581
1966	165261.7630	0.0106	1.9605	0.0632	0.9802	0.0000	719.9973
1967	164479.3249	0.0080	1.9524	0.0477	0.9762	0.0000	1098.8557
1968	164562.5049	0.0209	1.9527	0.1246	0.9764	0.0000	1307.8429
1969	162299.8302	0.0229	1.9260	0.1366	0.9630	0.0000	1258.9918
1970	160614.4780	0.0208	1.9052	0.1240	0.9526	0.0000	2015.4469
1971	159692.2871	0.0310	1.8962	0.1852	0.9481	0.0000	2587.1702
1972	157498.5901	0.0704	1.8698	0.4213	0.9349	0.0001	2816.6009
1973	149608.7709	0.0580	1.7773	0.3469	0.8886	0.0002	3475.1101
1974	146428.8658	0.0659	1.7405	0.3951	0.8703	0.0001	5728.0688
1975	143038.8172	0.0595	1.7003	0.3558	0.8502	0.0000	6450.9743
1976	141637.8055	0.0669	1.6837	0.4005	0.8418	0.0000	7263.1870
1977	139515.1826	0.0716	1.6583	0.4297	0.8292	0.0001	7573.4035
1978	137322.1993	0.0824	1.6316	0.4932	0.8158	0.0001	7990.1657
1979	134204.0899	0.0788	1.5949	0.4747	0.7975	0.0002	8423.5697
1980	132536.5545	0.0895	1.5744	0.5395	0.7872	0.0000	9074.1220
1981	129750.9307	0.0904	1.5412	0.5421	0.7706	0.0002	9393.5360
1982	127401.4285	0.0797	1.5176	0.4789	0.7588	0.0001	9906.9089
1983	127068.0184	0.0830	1.5145	0.4987	0.7573	0.0002	10207.6349
1984	126544.6172	0.1067	1.5073	0.6403	0.7537	0.0001	10224.9081
1985	122950.1494	0.1222	1.4659	0.7332	0.7330	0.0003	10401.9989
1986	118522.5328	0.1391	1.4133	0.8369	0.7066	0.0003	11092.9035
1987	115692.5767	0.1535	1.3762	0.9290	0.6881	0.0209	13948.5239
1988	113243.6721	0.1749	1.3485	1.0601	0.6743	0.0355	15672.3341
1989	107148.5992	0.1642	1.2768	1.0028	0.6384	0.0230	14226.0785
1990	101253.0375	0.1599	1.2092	0.9769	0.6046	0.0036	12283.6610
1991	95281.9882	0.1695	1.1395	1.0349	0.5698	-0.0157	10595.8679
1992	88158.7889	0.1611	1.0554	0.9835	0.5277	-0.0354	8970.4311
1993	89891.2784	0.1423	1.0769	0.8713	0.5385	0.0322	16404.7221
1994	93455.0258	0.1704	1.1149	1.0465	0.5575	0.0317	16207.9026
1995	89894.5800	0.1415	1.0738	0.8608	0.5369	0.0006	12816.4287
1996	92363.5049	0.1258	1.1054	0.7684	0.5527	0.0246	15431.2556
1997	97396.3781	0.1457	1.1663	0.8906	0.5831	0.0387	16746.9555
1998	98769.6755	0.1428	1.1844	0.8724	0.5922	0.0317	15848.7393
1999	98122.3448	0.1369	1.1763	0.8388	0.5881	0.0132	13801.0999
2000	97900.5990	0.1574	1.1733	0.9642	0.5867	0.0116	13687.3525
2001	94295.9818	0.1560	1.1307	0.9548	0.5654	-0.0020	12279.5081
2002	93194.6557	0.1373	1.1148	0.8367	0.5574	0.0102	13825.4360
2003	91925.5261	0.1681	1.1035	1.0274	0.5517	-0.0065	12054.4836
2004	89449.4892	0.1539	1.0728	0.9397	0.5364	0.0044	13457.7075
2005	92711.6024	0.1492	1.1125	0.9107	0.5563	0.0409	17234.8673
2006	97379.7602	0.1460	1.1675	0.8929	0.5838	0.0554	18747.2611
2007	100023.1271	0.1367	1.2057	0.8362	0.6028	0.0472	17555.2863
2008	102207.5097	0.1325	1.2315	0.8070	0.6158	0.0357	15977.0207
2009	100484.8411	0.1288	1.2123	0.7846	0.6061	0.0029	12312.0333
2010	102427.2989	0.1442	1.2329	0.8821	0.6165	0.0257	14844.6834
2011	98751.5409	0.1282	1.1836	0.7846	0.5918	-0.0060	11037.0861
2012	97869.8343	0.1132	1.1731	0.6921	0.5865	-0.0015	11920.3712
2013	98547.2487	0.1052	1.1797	0.6465	0.5898	-0.0048	11509.8764
2014	94744.0079	0.1208	1.1335	0.7418	0.5667	-0.0512	6769.5749
2015	88989.6070	0.1409	1.0631	0.8638	0.5315	-0.0662	5669.7691
2016	84864.8389	0.1527	1.0118	0.9412	0.5059	-0.0485	7959.0493
2017	79599.4083	0.1391	0.9483	0.8533	0.4742	-0.0585	7246.4580
2018	75964.7280	0.1186	0.9030	0.7296	0.4515	-0.0661	7018.5103

2019	74566.6793	0.1201	0.8891	0.7389	0.4445	-0.0639	7560.3539
2020	75039.8427	0.1151	0.8938	0.7088	0.4469	-0.0452	9072.2906
2021	74367.1861	0.1103	0.8860	0.6767	0.4430	-0.0599	7872.2408
2022	74527.7722	0.1064	0.8882	0.6546	0.4441	-0.0564	8217.8205
2023	76553.9634	0.1084	0.9150	0.6661	0.4575	-0.0373	9729.6355
2024	77942.4600	0.1165	0.9308	0.7142	0.4654	-0.0396	9452.2113

Table 6. Estimates of biomass (B), fishing mortality (F), biomass relative to B_{MSY} , and fishing mortality relative to F_{MSY} between 1950 and 2024 in the form of joint MCMC posteriors of JABBA model runs for the Pella-ASEM Main run.

	B	F	BBmsy	FFmsy	BB0	procB	SPt
1950	212483.7263	0.0027	2.7094	0.0163	0.9485	-0.0001	0.0000
1951	213647.9308	0.0027	2.7250	0.0161	0.9540	-0.0001	1756.5739
1952	214745.8938	0.0016	2.7385	0.0093	0.9587	0.0001	1618.5224
1953	215989.1558	0.0023	2.7530	0.0138	0.9638	-0.0001	1472.2927
1954	216966.1652	0.0021	2.7623	0.0124	0.9670	-0.0001	1247.3754
1955	217756.8596	0.0015	2.7701	0.0092	0.9697	0.0000	1156.6123
1956	218683.7118	0.0018	2.7787	0.0107	0.9728	-0.0001	1069.8263
1957	219375.6258	0.0029	2.7851	0.0175	0.9750	0.0000	995.7615
1958	219768.1220	0.0041	2.7870	0.0247	0.9757	-0.0001	861.0043
1959	219901.6682	0.0027	2.7852	0.0163	0.9750	0.0001	863.0191
1960	220297.4327	0.0023	2.7888	0.0139	0.9763	0.0001	884.1271
1961	220561.4172	0.0028	2.7919	0.0165	0.9774	0.0001	836.4112
1962	220709.5709	0.0032	2.7930	0.0190	0.9778	0.0000	770.9673
1963	220857.6257	0.0032	2.7933	0.0194	0.9779	0.0000	746.7091
1964	220895.9563	0.0036	2.7932	0.0216	0.9778	0.0000	758.9299
1965	220848.1809	0.0079	2.7918	0.0475	0.9774	0.0000	741.0150
1966	220012.8281	0.0079	2.7792	0.0475	0.9729	0.0001	779.6008
1967	219210.3097	0.0059	2.7679	0.0357	0.9690	0.0001	911.5806
1968	218928.2595	0.0156	2.7648	0.0938	0.9679	0.0001	1050.3382
1969	216624.0782	0.0171	2.7337	0.1025	0.9570	0.0002	1046.9603
1970	214287.9866	0.0156	2.7047	0.0932	0.9469	0.0001	1404.3354
1971	212651.3777	0.0233	2.6842	0.1395	0.9397	-0.0001	1731.0340
1972	209519.4654	0.0530	2.6451	0.3183	0.9260	0.0000	1950.7548
1973	200569.5402	0.0432	2.5334	0.2600	0.8869	0.0001	2370.2581
1974	195451.7782	0.0496	2.4676	0.2976	0.8638	0.0000	3624.3491
1975	189785.1408	0.0448	2.3970	0.2689	0.8391	-0.0002	4277.5748
1976	186166.8878	0.0515	2.3501	0.3085	0.8227	-0.0001	4985.0335
1977	181905.7132	0.0553	2.2967	0.3324	0.8040	-0.0001	5416.5362
1978	177564.2906	0.0639	2.2434	0.3838	0.7854	-0.0001	5960.0984
1979	172442.1556	0.0614	2.1795	0.3701	0.7630	-0.0001	6447.9079
1980	168712.3062	0.0703	2.1336	0.4231	0.7469	-0.0001	7069.9485
1981	164098.2755	0.0714	2.0747	0.4298	0.7263	0.0000	7497.6858
1982	160241.6390	0.0634	2.0264	0.3816	0.7094	0.0000	8004.0426
1983	158298.1400	0.0672	2.0017	0.4021	0.7008	0.0001	8433.1893
1984	155981.8677	0.0866	1.9748	0.5205	0.6913	0.0002	8695.7313
1985	150903.0179	0.1000	1.9119	0.6030	0.6693	0.0002	8935.5803
1986	144789.8973	0.1147	1.8346	0.6895	0.6422	0.0002	9441.3063
1987	138465.7323	0.1300	1.7529	0.7836	0.6136	0.0090	10977.4355
1988	133085.0372	0.1501	1.6836	0.9059	0.5894	0.0219	13243.2111
1989	124741.3517	0.1421	1.5813	0.8566	0.5536	0.0136	12418.2147
1990	117285.3439	0.1380	1.4886	0.8343	0.5211	-0.0016	10667.0647
1991	109985.8541	0.1463	1.3985	0.8863	0.4896	-0.0168	9306.2522
1992	101862.6318	0.1396	1.2941	0.8416	0.4530	-0.0316	8117.5724

1993	103258.9548	0.1245	1.3133	0.7515	0.4597	0.0338	16027.6866
1994	106322.9073	0.1491	1.3526	0.8994	0.4735	0.0352	16075.6140
1995	102952.2719	0.1229	1.3100	0.7412	0.4586	0.0068	12756.6920
1996	105586.1256	0.1099	1.3444	0.6646	0.4706	0.0310	15621.0745
1997	111031.0328	0.1280	1.4129	0.7728	0.4946	0.0435	17026.4184
1998	112425.2245	0.1255	1.4288	0.7590	0.5002	0.0350	16054.3393
1999	111711.2969	0.1205	1.4192	0.7283	0.4968	0.0162	13652.4909
2000	111270.7991	0.1382	1.4176	0.8356	0.4963	0.0154	13615.0465
2001	107579.9963	0.1369	1.3694	0.8274	0.4794	0.0030	12026.9362
2002	105973.0749	0.1198	1.3471	0.7260	0.4716	0.0139	13454.6700
2003	104981.5626	0.1462	1.3386	0.8826	0.4686	0.0012	12084.7444
2004	102547.5819	0.1340	1.3074	0.8088	0.4577	0.0111	13347.6462
2005	105909.7018	0.1307	1.3478	0.7884	0.4718	0.0468	17585.4931
2006	110811.5143	0.1281	1.4119	0.7737	0.4943	0.0614	19232.6339
2007	114343.9191	0.1199	1.4549	0.7237	0.5093	0.0517	18116.0070
2008	116623.3797	0.1159	1.4827	0.7002	0.5191	0.0382	16328.0500
2009	114978.7001	0.1122	1.4642	0.6801	0.5126	0.0068	12275.1439
2010	116819.4050	0.1265	1.4853	0.7630	0.5200	0.0272	14790.0072
2011	112548.6147	0.1121	1.4341	0.6777	0.5020	-0.0047	10775.5018
2012	111138.3609	0.0991	1.4178	0.5998	0.4963	-0.0007	11489.0518
2013	111729.6235	0.0921	1.4206	0.5572	0.4973	-0.0033	11255.2635
2014	107455.7239	0.1060	1.3683	0.6404	0.4790	-0.0478	6232.9455
2015	101249.1988	0.1228	1.2898	0.7403	0.4515	-0.0618	4956.0382
2016	96338.0326	0.1330	1.2260	0.8070	0.4292	-0.0448	7226.6256
2017	90615.6162	0.1208	1.1539	0.7298	0.4040	-0.0519	6831.0199
2018	86472.5249	0.1033	1.1018	0.6251	0.3857	-0.0604	6575.3379
2019	85069.6502	0.1048	1.0829	0.6329	0.3791	-0.0570	7205.0333
2020	85189.4377	0.1007	1.0867	0.6077	0.3804	-0.0390	8929.5363
2021	84581.9093	0.0959	1.0778	0.5802	0.3773	-0.0537	7670.4216
2022	84838.6060	0.0931	1.0823	0.5636	0.3789	-0.0481	8194.0005
2023	86657.3319	0.0953	1.1082	0.5754	0.3880	-0.0310	9704.7528
2024	87867.4972	0.1034	1.1215	0.6223	0.3926	-0.0335	9356.7510

Table 7. Estimates of biomass (B), fishing mortality (F), biomass relative to B_{MSY} , and fishing mortality relative to F_{MSY} between 1950 and 2024 in the form of joint MCMC posteriors of JABBA model runs for the Pella-m Main run.

	B	F	BBmsy	FFmsy	BB0	procB	SPt
1950	229090.9091	0.0025	2.7096	0.0156	0.9506	-0.0001	0.0000
1951	229819.9606	0.0025	2.7203	0.0154	0.9560	0.0001	1695.2755
1952	230447.8871	0.0015	2.7309	0.0089	0.9601	0.0001	1518.3351
1953	231155.7172	0.0021	2.7423	0.0132	0.9652	0.0001	1380.5692
1954	231878.9561	0.0019	2.7502	0.0118	0.9686	0.0000	1208.1509
1955	232544.1202	0.0015	2.7575	0.0089	0.9716	0.0001	1139.7267
1956	233128.5886	0.0017	2.7660	0.0103	0.9743	0.0000	1002.8826
1957	233622.2053	0.0027	2.7724	0.0168	0.9768	0.0000	907.6052
1958	233601.7284	0.0039	2.7749	0.0238	0.9777	0.0000	797.3849
1959	233507.7204	0.0026	2.7740	0.0157	0.9772	0.0001	807.1379
1960	233558.0093	0.0022	2.7772	0.0134	0.9781	0.0001	827.8936
1961	233788.0333	0.0026	2.7801	0.0159	0.9791	0.0001	806.7109
1962	233971.8564	0.0030	2.7812	0.0183	0.9798	0.0001	752.2294
1963	233976.0827	0.0031	2.7814	0.0188	0.9798	0.0000	727.1924
1964	234047.5578	0.0034	2.7804	0.0207	0.9797	0.0001	736.7469
1965	234014.7122	0.0075	2.7793	0.0456	0.9794	0.0000	703.4851
1966	233071.4774	0.0075	2.7676	0.0456	0.9748	0.0000	693.7180

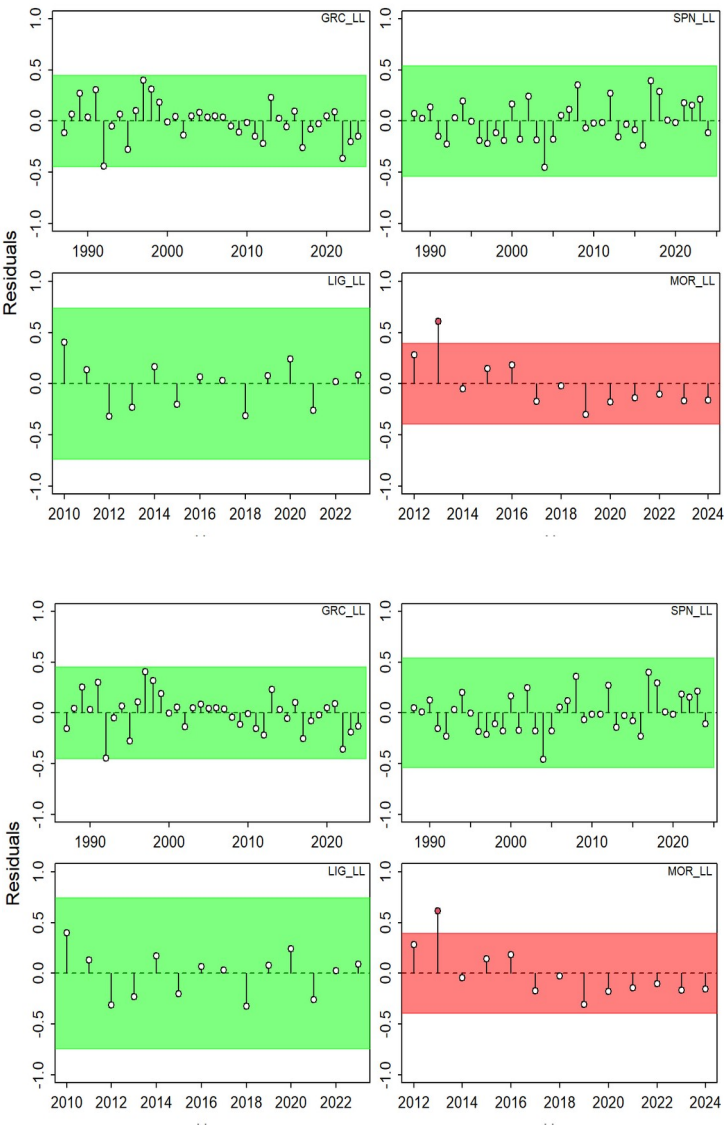
1967	232214.6801	0.0056	2.7566	0.0345	0.9711	0.0000	869.5117
1968	231830.4293	0.0148	2.7526	0.0903	0.9700	0.0000	1013.4710
1969	229525.0554	0.0161	2.7236	0.0987	0.9596	0.0001	1031.5563
1970	227190.9232	0.0146	2.6958	0.0896	0.9497	-0.0001	1373.0068
1971	225406.9830	0.0219	2.6750	0.1343	0.9428	0.0000	1741.5278
1972	222272.9845	0.0499	2.6391	0.3060	0.9301	0.0000	1946.6712
1973	213558.9743	0.0407	2.5306	0.2497	0.8928	0.0001	2341.7830
1974	208222.7140	0.0462	2.4673	0.2853	0.8711	0.0001	3605.7467
1975	202660.5377	0.0417	2.4020	0.2572	0.8480	0.0001	4293.1803
1976	198989.0928	0.0475	2.3595	0.2930	0.8330	0.0001	5050.9227
1977	194997.1154	0.0512	2.3080	0.3156	0.8157	0.0000	5527.9708
1978	190698.4711	0.0587	2.2563	0.3616	0.7982	0.0002	6014.6055
1979	185715.7725	0.0570	2.1959	0.3527	0.7775	0.0002	6614.9005
1980	182045.6251	0.0652	2.1510	0.4015	0.7626	0.0001	7086.6153
1981	177411.4452	0.0658	2.0966	0.4071	0.7429	0.0003	7559.2489
1982	173141.1527	0.0584	2.0482	0.3622	0.7260	0.0002	8057.8447
1983	171384.9437	0.0620	2.0246	0.3835	0.7180	0.0002	8426.7373
1984	169005.8516	0.0802	1.9980	0.4962	0.7088	0.0003	8606.0647
1985	163843.4921	0.0924	1.9363	0.5714	0.6872	0.0001	8855.2563
1986	157714.5426	0.1060	1.8605	0.6547	0.6609	0.0004	9438.3724
1987	151064.4511	0.1190	1.7862	0.7399	0.6339	0.0086	11001.5904
1988	145646.6458	0.1382	1.7249	0.8581	0.6109	0.0207	13317.9185
1989	137075.9876	0.1298	1.6251	0.8101	0.5756	0.0122	12300.3798
1990	129195.5043	0.1259	1.5343	0.7871	0.5418	-0.0035	10401.7976
1991	121666.6583	0.1327	1.4434	0.8304	0.5108	-0.0183	9039.9001
1992	113113.8875	0.1268	1.3391	0.7924	0.4748	-0.0318	7743.5819
1993	114744.6854	0.1125	1.3577	0.7088	0.4809	0.0298	16013.2243
1994	117518.4762	0.1359	1.3956	0.8536	0.4942	0.0314	16128.2918
1995	114157.1701	0.1114	1.3516	0.6999	0.4783	0.0046	12642.3884
1996	116860.3548	0.1000	1.3859	0.6274	0.4907	0.0286	15679.1926
1997	122158.9335	0.1166	1.4538	0.7276	0.5134	0.0404	17175.2373
1998	123868.8945	0.1144	1.4748	0.7118	0.5199	0.0328	16052.0406
1999	122857.9449	0.1098	1.4664	0.6886	0.5168	0.0150	13719.7465
2000	122703.3988	0.1261	1.4627	0.7891	0.5160	0.0137	13608.9512
2001	118972.7008	0.1238	1.4156	0.7777	0.4999	0.0015	11913.0351
2002	117411.7161	0.1086	1.3952	0.6779	0.4923	0.0116	13552.2533
2003	116233.5808	0.1334	1.3817	0.8364	0.4885	-0.0006	11940.9594
2004	113588.9049	0.1219	1.3520	0.7626	0.4775	0.0095	13307.1021
2005	117162.4187	0.1185	1.3959	0.7409	0.4929	0.0437	17731.1134
2006	122476.8584	0.1168	1.4603	0.7315	0.5154	0.0566	19398.7893
2007	126030.8201	0.1096	1.5031	0.6834	0.5307	0.0485	18256.4326
2008	128426.0364	0.1054	1.5306	0.6586	0.5404	0.0363	16412.1860
2009	126795.4082	0.1023	1.5117	0.6380	0.5325	0.0061	12258.5312
2010	128532.1571	0.1146	1.5350	0.7168	0.5417	0.0261	14952.0619
2011	124266.0478	0.1012	1.4811	0.6343	0.5225	-0.0049	10613.3508
2012	123102.1595	0.0896	1.4684	0.5623	0.5180	-0.0007	11527.8645
2013	123142.0950	0.0838	1.4705	0.5245	0.5182	-0.0055	10887.4038
2014	118579.6545	0.0956	1.4145	0.5978	0.4981	-0.0461	5820.6217
2015	111664.9541	0.1111	1.3319	0.6973	0.4695	-0.0605	4504.0844
2016	106330.9358	0.1205	1.2676	0.7563	0.4476	-0.0436	7015.7551
2017	100264.2069	0.1084	1.1952	0.6805	0.4221	-0.0522	6427.5065
2018	96037.7505	0.0931	1.1430	0.5867	0.4034	-0.0593	6304.9154
2019	94337.8984	0.0944	1.1229	0.5932	0.3966	-0.0552	7115.3574
2020	94607.9458	0.0907	1.1283	0.5693	0.3975	-0.0390	8837.2387
2021	93651.0919	0.0864	1.1169	0.5421	0.3939	-0.0515	7644.7243
2022	93818.5256	0.0839	1.1179	0.5273	0.3950	-0.0464	8147.8448

2023	96110.3080	0.0859	1.1461	0.5362	0.4044	-0.0292	9770.9399
2024	97341.0234	0.0932	1.1640	0.5811	0.4104	-0.0321	9394.1292

2.3 Model Diagnostics

2.3.1 Residuals: Runs tests and RMSE

Runs test results (Figure 13) show no evidence of temporal autocorrelation in the residuals of the abundance index fits, with the exception of one Morocco longline index. This particular index consistently fails the runs test across the main models, though only marginally. The 2018 residual is small and near zero, and it is crucial for the passing of this test. A slight change in CPUE fits could make the 2018 MOR_LL residuals be small and positive and pass the runs test.



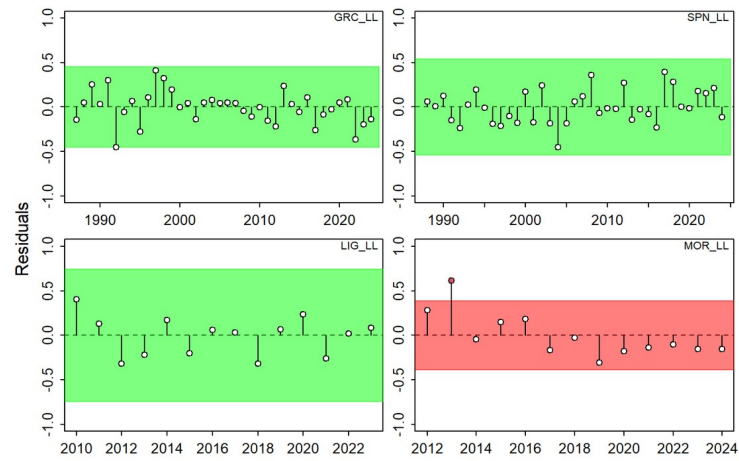
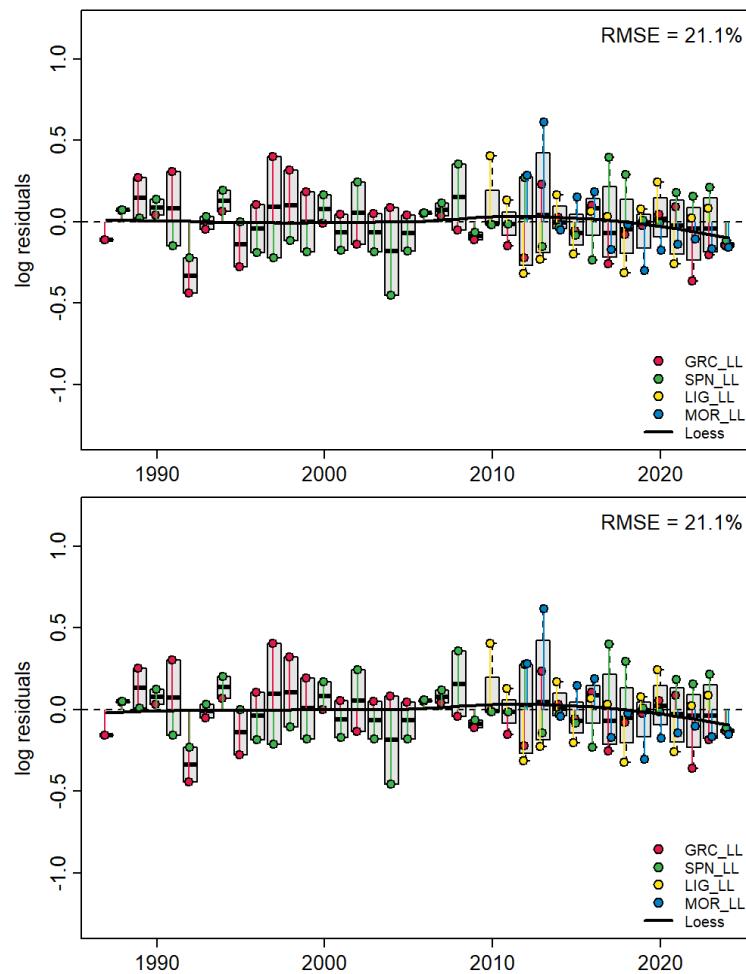


Figure 13. Run tests to evaluate the randomness of the residuals of the CPUE series by fleet. Green panels indicate no evidence of lack of randomness of time-series residuals while red panels indicate the opposite. The inner shaded area shows three standard errors from the overall mean and red circles identify a specific year with residuals greater than this threshold value (3x sigma rule). Top panel: Schaefer Main run; Middle panel: Pella-Tollimson ASEM Main run; Bottom panel: Pella-est_m Main run.

The overall goodness-of-fit was very similar across the three models, achieving a value of $RMSE = 21.1\%$ (**Figure 14**). The global tendency of all the residuals is near 0 for most of the model years. However, after 2020 there appears to be a tendency towards negative residuals.



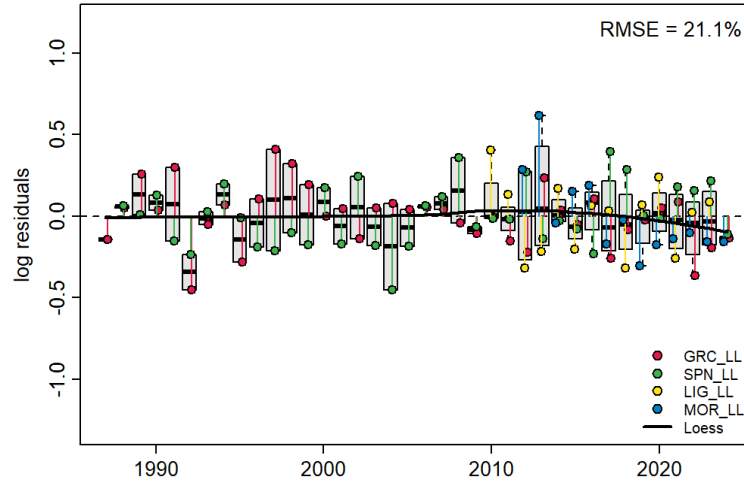


Figure 14. Residual diagnostic plots. Boxplots indicate the median and quantile for all residuals by year and fleet. Solid black line represents a loess smoother. Top panel: Schaefer Main run; Middle panel: Pella-Tollimson ASEM Main run; Bottom panel: Pella-est_m Main run.

MCMC draws: stability tests and traces

The MCMC convergence diagnostics proposed by Heidelberg and Welch (Heidelberg and Welch, 1992) and Geweke (1992) were successfully passed by all estimated key parameters in each of the Main run models (see **Table 3**). Visual examination of the MCMC trace plots further confirmed satisfactory convergence.

MCMC traces (**Figure 15**) and stability statistical tests support that drawn samples from the MCMC process come from stationary distributions, which is a signal of convergence of the models. Fits to the standardized CPUE indices generally fall within the total observational uncertainty values.

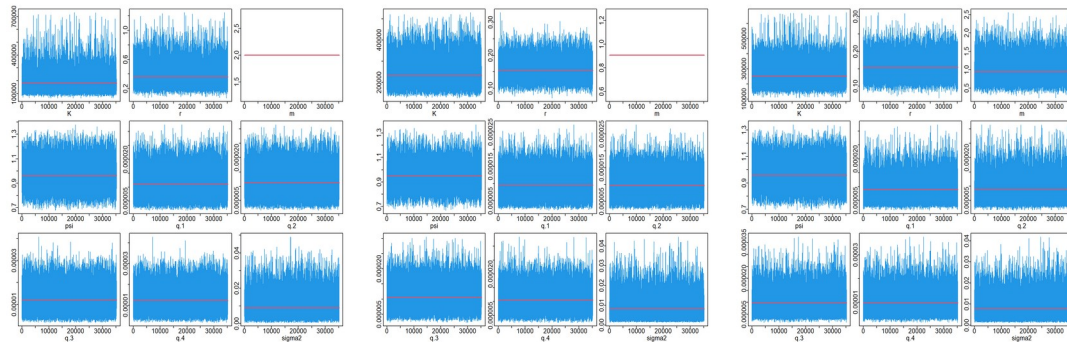
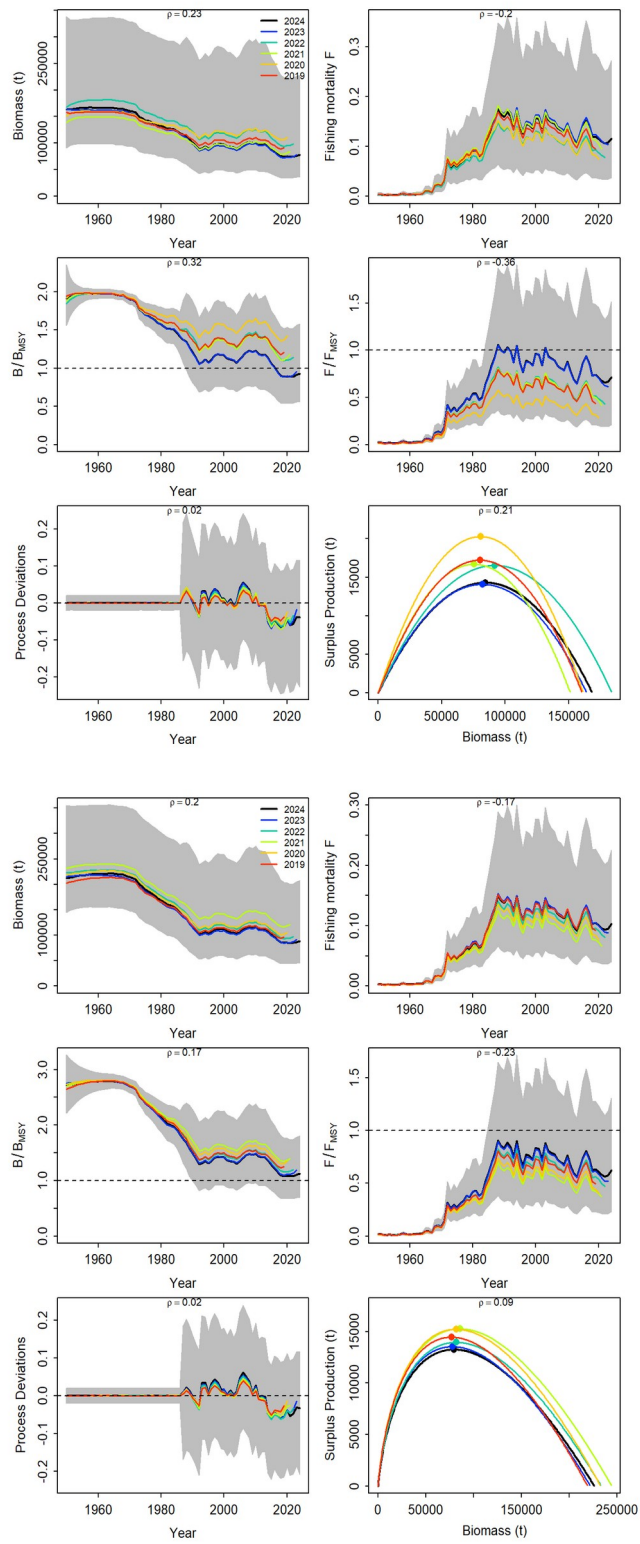


Figure 15. MCMC diagnostic plots for the JABBA model, showing posterior draws, stability diagnostics, and trace plots used to assess convergence and mixing of the Markov chains.

Retrospective runs

Retrospective runs were produced in order to assess model stability with respect to the last 5 years of data with respect to 2024. Summary plots of all retrospective runs on the Main models are displayed in **Figure 16**. All of the main runs display a notable retrospective pattern indicating either a strong influence of the final years of data or model instability associated to model misspecification or conflicting data sources. For longer lived species, the reference range of acceptable Mohn's rho value is [-0.15, 0.20] (Carvalho, 2021). In the case of the Main runs presented, the closest one to fall within this range is the Pella-est-m model.

Using the same retrospective runs, hindcast cross-validation indicators were computed. MASE scores rank predictive ability of the different CPUE series consistently across models (**Figure 17**).



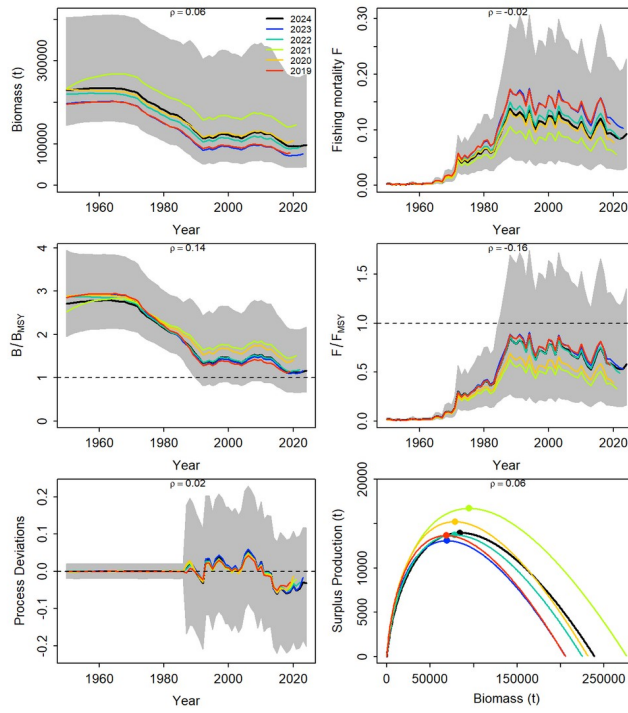
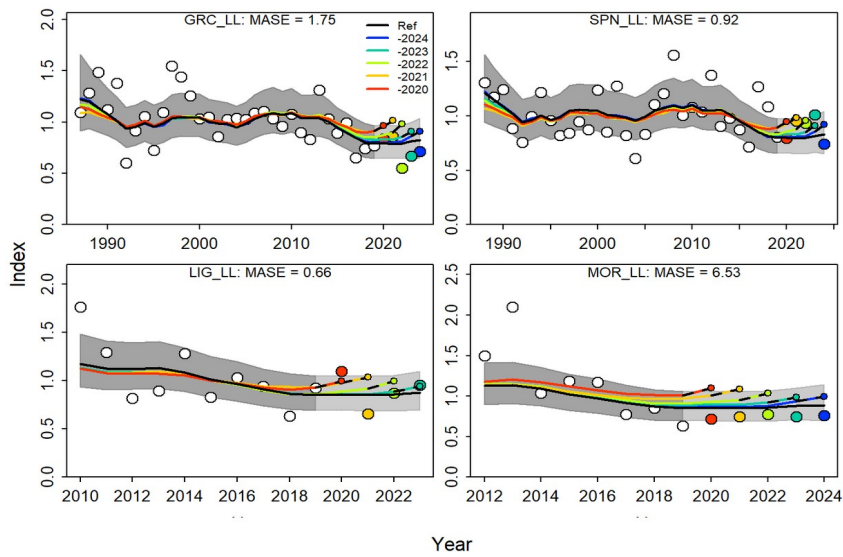


Figure 16. Retrospective analysis performed by removing one year at a time sequentially ($n=5$) and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario. Top panel: Schaefer Main run; Middle panel: Pella-Tollimson ASEM Main run; Bottom panel: Pella-est_m Main run.



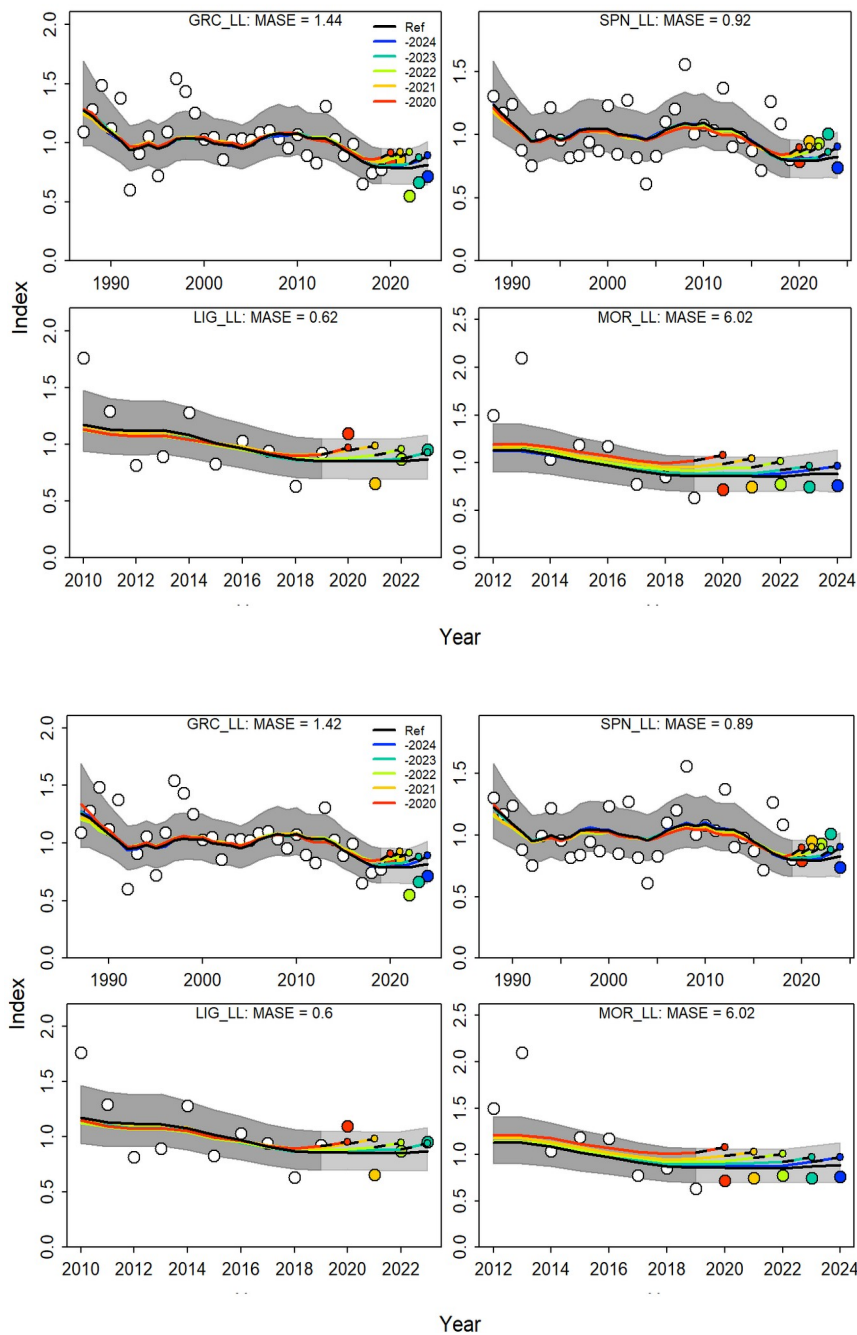


Figure 17. Hindcast analysis for the JABBA model. Biomass trajectories from sequentially truncated assessments are compared with the full assessment to evaluate predictive performance; forecast accuracy is summarized using the mean absolute scaled error (MASE). Top panel: Schaefer Main run; Middle panel: Pella-Tollimson ASEM Main run; Bottom panel: Pella-est_m Main run.

2.4 Sensitivity runs

(i) Jackknife CPUE

The Jackknife sensitivity analyses on the CPUE indices showed consistent stock trajectory trends, with sensible differences in scale and productivity as seen in **Figure 18**. The Jackknife Pella-Tomlinson runs where the Greece longline index is not fitted stand out with an extremely high estimated fishing mortality during the late 1990s. However, after further examining these runs, it was noted that posterior fits of key parameters display bimodality, a sign of convergence issues. In that sense, GRC_LL can be seen as stabilizing series for the model. Differences between the two Pella-Tomlinson runs become more apparent in this sensitivity comparison. This is because Pella-est_m estimates both parameters of the production curve, while the Pella ASEM model has fixed shape.

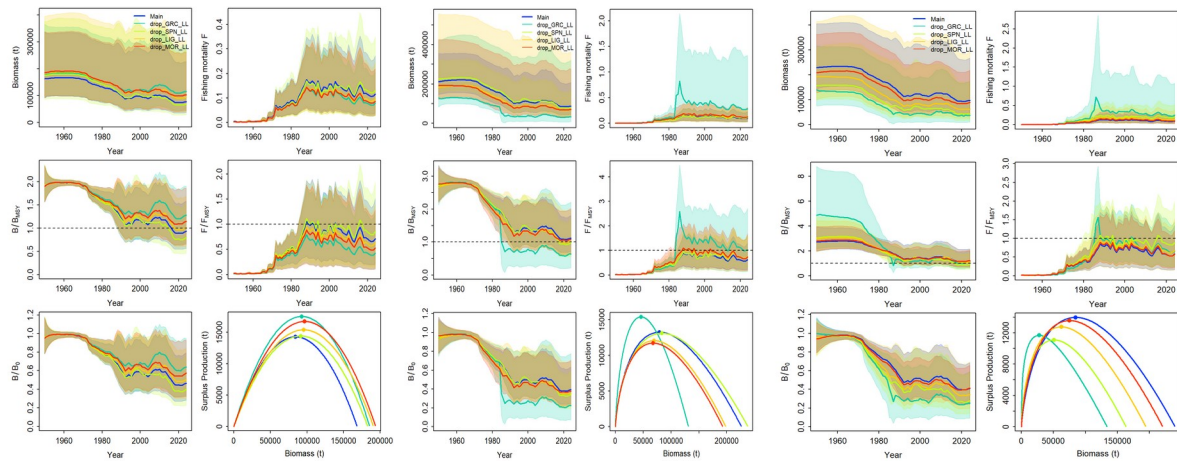


Figure 18. Jackknife sensitivity analyses on the CPUE indices. Right panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Left panel: Pella-est_m Main run.

(ii) Scenarios of catch under-reporting 1950-1986

Figure 19 shows the different scenarios of historic under-reported catches that were tested. The catch time series for years 1950 to 1986 is scaled with four different multipliers (x1.2, x1.5, x2, x3) accounting for values that range from moderate to high levels of possible under-reporting.

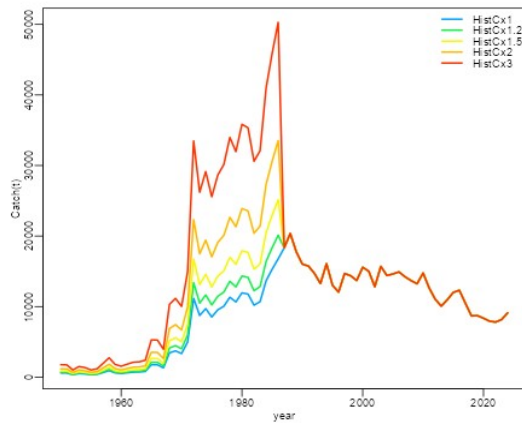


Figure 19. Different sensitivity scenarios on under-reporting of catches from 1950 to 1986. HistCx1: no under-reporting on catches; HistCx1.2: historical catches multiplied by 1.2; HistCx1.5: historical catches multiplied by 1.5. HistCx2: historical catches multiplied by 2. HistCx3: historical catches multiplied by 3.

The sensitivity runs associated to various levels of under-reporting of catches prior to 1987 generally showed steeper declines in B and increases in F with higher levels of historic under-reporting of catches, as well as higher K and MSY (**Figure 20**) for the three Main models tested, specially in those scenarios considering the higher historic under-reporting of catches (HistCx2 and HistCx3).

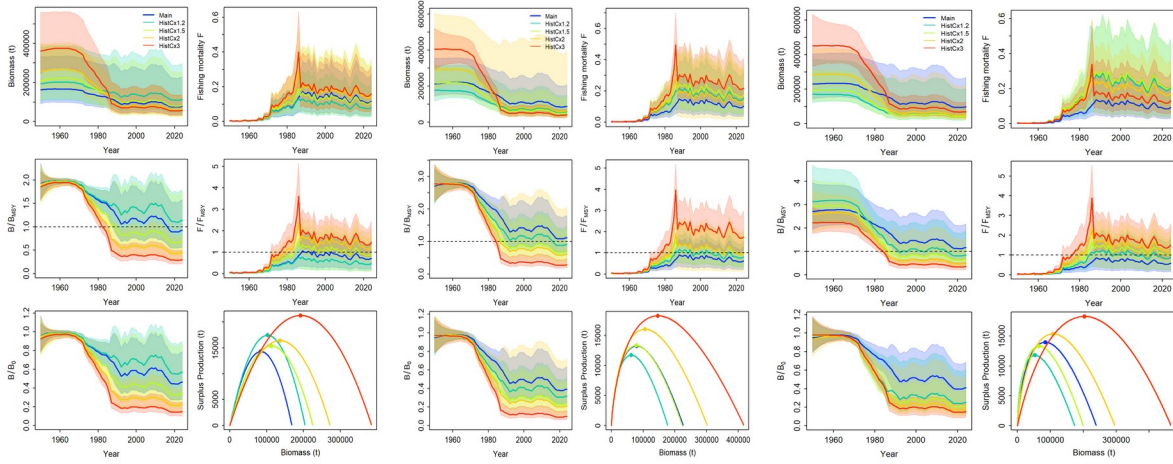


Figure 20. Summary of the main outcomes of the sensitivity analyses on under-reporting of historical catches. Left panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Right panel: Pella-est_m Main run. HistCx1: no under-reporting on catches; HistCx1.2: historical catches multiplied by 1.2; HistCx1.5: historical catches multiplied by 1.5. HistCx2: historical catches multiplied by 2. HistCx3: historical catches multiplied by 3.

(iii) Initial depletion prior centered at 0.85

The sensitivity analysis considering a lower estimate of the initial depletion ($\phi = B_{1950}/K$) showed a minor impact on the three Main models (**Figure 21**). It showed slightly higher trends in B , as well as lower F and higher K and MSY for the Schaefer and Pella-ASEM model configurations. The Pella-Est_m shows the converse result. A noteworthy aspect of this sensitivity run is the increasing trend displayed by the estimated dynamics in the first years of the lower- ϕ model (1950 – 1960). This trend is most noticeable in relative scales (B/B_{MSY} and B/B_0 panels in **Figure 21**). In all three cases of the lower- ϕ sensitivity run, biomass attains B_0 levels around 1960.

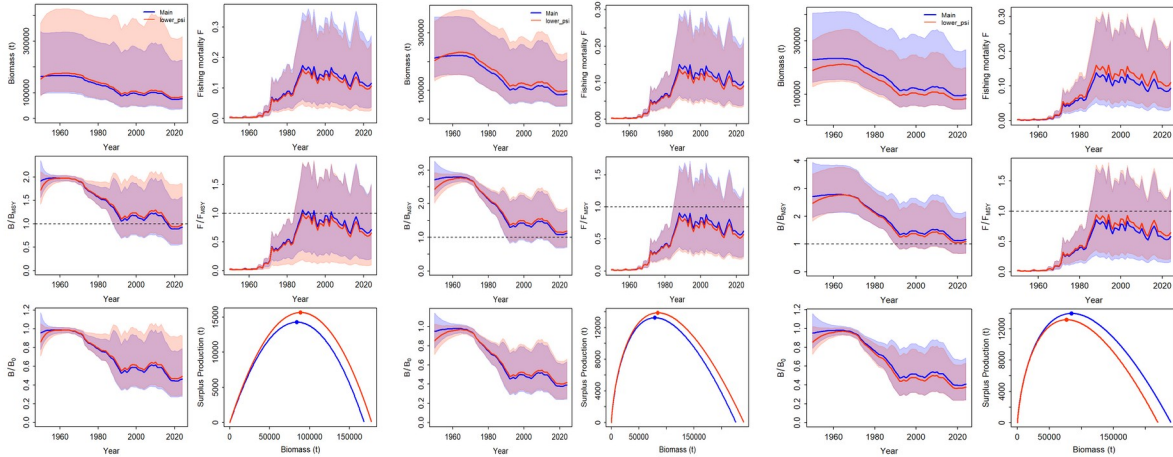


Figure 21. Summary of the main outcomes of the sensitivity analyses on lower initial depletion value. Left panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Right panel: Pella-est_m Main run.

(iv) No CV on catches

The sensitivity analysis consisting on removing error on catches showed little effect both, on the Schaefer and the Pella-est_m production models, whereas it had a higher impact on the Pella-ASEM model configuration (**Figure 22**). In contrast, on the Pella-ASEM model removing the error on catches showed higher scale on the trajectories of B , as well as lower F , in addition to higher K and MSY .

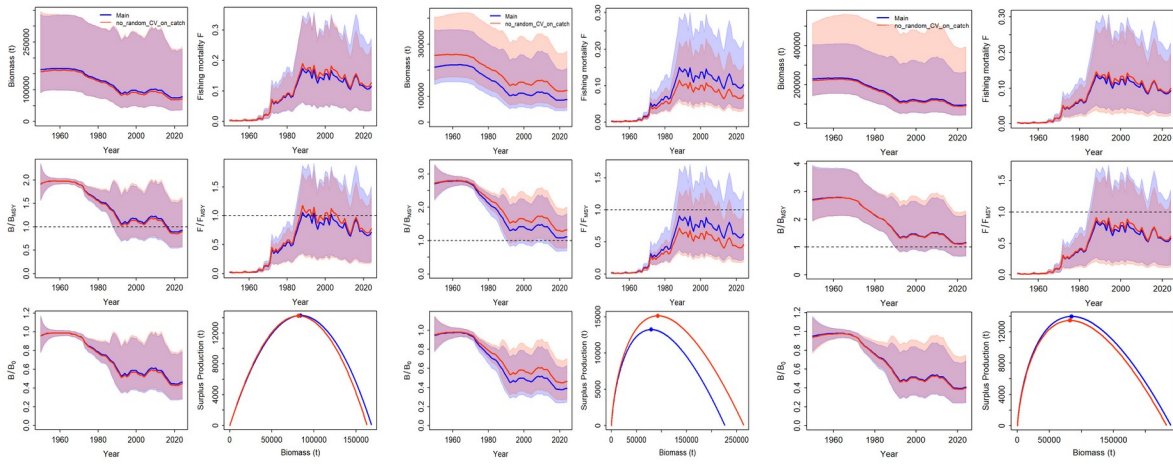


Figure 22. Summary of the main outcomes of the sensitivity analyses on removing the error on catches. Left panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Right panel: Pella-est_m Main run.

(v) Include historical abundance indices

The use of the historical indices of abundance (Sicilian gillnet and early Ligurian longline) results in the estimation of a less productive stock and this is reflected consistently in the population dynamics with lower biomass and higher fishing mortality, both in relative and absolute terms (**Figure 23**). It is worth to mention that the latter years of the Sicilian Gillnet index (2006 onwards) are not fitted by the estimated abundance series. Contrarily to main runs, which display two distinct ‘phases’ of process deviations (non-negative in approximately 1990-2010 and negative after 2010), in the historical abundance indices sensitivity runs process deviations oscillate more around 0 during the mentioned periods.

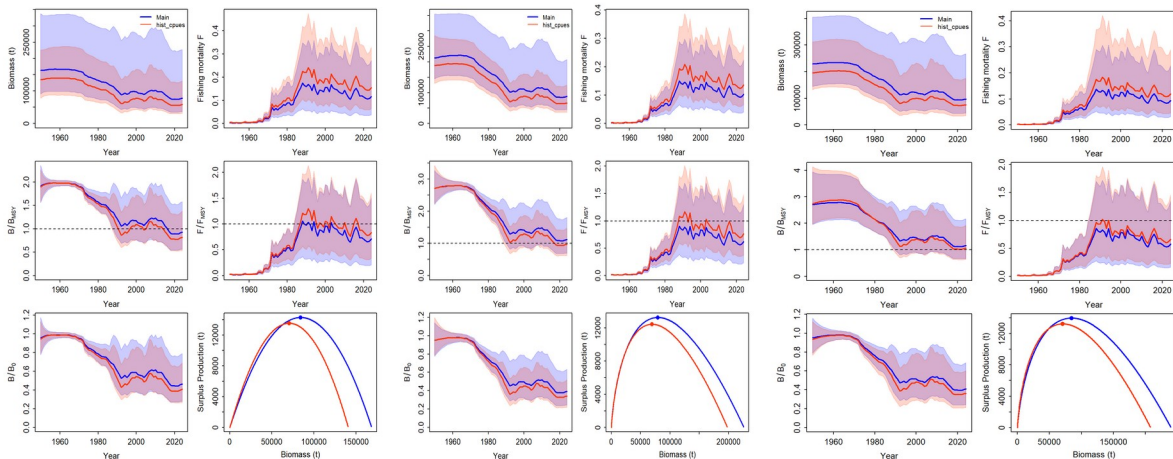


Figure 23. Sensitivity analysis on the use of historical CPUE indices. Left panel: Schaefer Main run; Central panel: Pella-Tollimson ASEM Main run; Right panel: Pella-est_m Main run.

3. Discussion

Three main models are proposed as reference cases for estimating Mediterranean swordfish stock status. Data inputs, configurations and sensitivity runs were agreed during the Data Preparatory Meeting (Anon., 2026). The main model outcomes provide a different perspective on the stock status with respect to the 2020 Stock assessment (Anon., 2020). The update on the standardized CPUE indices has been identified as a potential driver of this shift.

The three main models show similar trends and overall results. However, there is a sensible difference in scale between the Schaefer model and the Pella-Tomlinson ones. Productivity is estimated to be very similar between the Pella-Tomlinson fixed m (ASEM) and the Pella-Tomlinson estimated m approach.

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References

- Anon. 2015. Report of the 2014 ICCAT Mediterranean Swordfish Stock Assessment meeting (Heraklion, Greece, 21-25 July 2014). Collect. Vol. Sci. Pap. ICCAT, 71 (5): 1870-1979.
- Anon. 2020. Report of the 2020 ICCAT Mediterranean swordfish stock assessment meeting (Online, 25 May - 2 June 2020). Collect. Vol. Sci. Pap. ICCAT, 77(3): 1-78 (2020).
- Anon. 2026. Report of the Atlantic Swordfish Data Preparatory meeting (Hybrid, Madrid, Spain, 23-27 March). Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/004: 1-97 (2026)
- Borrego-Santos R., Puerto M.A, Alegría A, Rueda L, Rodríguez E, Gómez-Vives MJ, Moreno J, Carrera M, Saber S, Poisson F, and Macías D. Updated analysis of maturity in swordfish (*Xiphias gladius*) from western Mediterranean Sea: From 2018 to 2025. Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/055: 1-14 (2026)
- Garibaldi, F., and Tserpes, G., 2014. Standardization of swordfish catch rates from the Ligurian surface drifting longline fisheries for the period 1991-2009. SCRS/2014/112
- Garibaldi F, Bottero F, Lanteri L, and Pinto C. Updated standardized swordfish catch rates from longline fisheries operating in the Ligurian Sea (Italy, western Mediterranean) (period 1990–2024). Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/042: 1-13 (2026).
- Geweke, J., 1992. Evaluating the accuracy of sampling-based approaches to the calculation of posterior moments., in: Berger, J.O., Bernardo, J.M., Dawid, A.P., Smith, A.F.M. (Eds.), Bayesian Statistics 4: Proceedings of the Fourth Valencia International Meeting. Clarendon Press, Oxford, pp. 169–193.
- Heidelberger, P., Welch, P.D., 1992. Simulation run length control in the presence of an initial transient. Oper. Res. 31, 1109–1144. <https://doi.org/10.1287/opre.31.6.1109>
- Jardim, E., Millar, C.P., Mosqueira, I., Scott, F., Osio, G.C., Ferretti, M., Alzorriz, N., Orio, A., 2015. What if stock assessment is as simple as a linear model? The a4a initiative. ICES J. Mar. Sci. 72, 232–236.
- Kell, L.T., Mosqueira, I., Grosjean, P., Fromentin, J., Garcia, D., Hillary, R., Jardim, E., Mardle, S., Pastoors, M.A., Poos, J.J., Scott, F., Scott, R.D., 2007. FLR: an open-source framework for the evaluation and development of management strategies. ICES J. Mar. Sci. 64, 640–646.
- Layachi M., Ikkiss A, Abid N, Serghini M, and Bensbai J. Updated catch rates of swordfish (*Xiphias gladius*) caught by Moroccan longline fleet in the Mediterranean Sea, 2012–2024. Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/048: 1-7 (2026).
- Ortiz, M and Kimoto, A. 2026. Estimation of undersize Mediterranean swordfish (*Xiphias gladius*) dead discards between 2019-2024 for the main longline fleets. SCRS/2026/090
- R Core Team 2026. *_R: A Language and Environment for Statistical Computing_*. R Foundation for Statistical Computing, Vienna, Austria.<https://www.R-project.org/>>
- Rueda L., Moreno, J.F., Ortiz de Urbina, J.M., Puerto, M.A. and Macías, D. 2026. Swordfish discard estimates in the Spanish Mediterranean longline fleet using the Bycatch Estimator Tool. Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/046: 1-9 (2026).
- Saber, S., Rueda, L., Macías, D., Garcia, S., Rioja, P., Puerto, M.A., Ortiz de Urbina, J., de la Serna, J.M. 2026. Standardized catch rates for Mediterranean swordfish from the Spanish longline fishery. 1988-2024. SCRS/2026/029.
- Tserpes, G., Peristeraki, P., Di Natale A. and Mangano, A. Analysis of swordfish (*Xiphias gladius*) catch rates in the central-eastern Mediterranean. SCRS/2010/085
- Tserpes G., Peristeraki P. CPUE trends of the Greek surface longline swordfish fishery in the eastern Mediterranean. Col. Vol. Sci. Pap. ICCAT, 83(2), SCRS/2026/039: 1-6 (2026)

- Winker, H., Carvalho, F., Kapur, M., 2018. JABBA: Just Another Bayesian Biomass Assessment. Fish. Res. 204, 275–288. <https://doi.org/http://doi.org/10.1016/j.fishres.2018.03.01>
- Winker, H., Mourato, B., Chang, Y. 2020a. Unifying parameterizations between age-structured and surplus production models: an application to Atlantic white marlin (*kajikia albida*) with simulation testing. Collect. Vol. Sci. Pap. ICCAT, 76(4): 219-234.
- Winker H., Kimoto A., Mourato B.L., Tserpses G., and Ortiz M. 2020b. Development of Bayesian state-space surplus production model JABBA for assessing the Mediterranean swordfish (*Xiphias gladius*) stock. Collect. Vol. Sci. Pap. ICCAT, 77(3): 508-536 (2020).
- Winker H, Carvalho F, Kapur M (2026). _JABBA: Just Another Bayesian Biomass Assessment_. R package version 2.3.5, commit bc14cb1233c1b5cf103c9e21c181ba81c2763d4c, <<https://github.com/jabbamodel/JABBA>>.